

LightSpeed 4.X

Pre-Installation Manual

This manual applies only to install-based LightSpeed 4.X systems. It is not effective for systems shipped in 2006 or later by GoldSeal.



Effectivity

The information in this document applies to the LightSpeed16 (w/16-slice MDAS) CT Scanner.



LEGAL NOTES

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End of Legal Notes

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu áánleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS

(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploataavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploataavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION



Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION



Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the Healthcare PQR Process to report all omissions, errors, and defects in this publication.

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Revision History

Revision	Date	Reason for change
0	08/19/02	New Product Introduction
1	03/12/03	Chapter 9: Corrected console cable information (in Table 9-3) Added information on Global Console - Linux
2	01/13/04	CQA 13009316: Updated Cooling Requirements in Chapter 5. General updates.
3	05/18/04	CTCge82492: Corrected Gantry Floor Loading weight/area figure. CTCge87943: Updated Floor Anchor Information. Chapter 9: Added "CT Systems - Partial UPS Back-up Options" table
4	12/08/04	Floor Anchor updates.
5	03/06/07	Added NGPDU-2 information to support some CT and CT PET systems shipped in 2006 or later.
6	08/31/07	Replaced front and back covers with new formats; replaced warnings with updated information. SPR FCTge29822, PQR 13111959: Revision of Regulatory information to meet new standards and requirements; metric measurements added for all measurements.
7	03/31/08	Removed information added in revisions 5 and 6 to restore the manual to installation-based status. Maintained updated covers and warnings.

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Typical UPS

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Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task. Please become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION

**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0

Conventions Used For General Information

2.1 Document Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

2.2 Page Layout

Headers and footers in this publication are designed to allow you to quickly identify your location. The document’s part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

Section 3.0

Computer Screen Output and Input

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: This paragraph denotes computer screen fixed output. It’s output is fixed from the sense that it does not vary from application to application.

Example: *This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example:*
<variable_ouput>

Example: This paragraph denotes fixed input. It’s typed input that will not vary from application to application.

Example: *This paragraph denotes computer input that can vary from application to application. Variable input will normally be found placed between greater than and lesser than operators. For example: <variable_input>*

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Chapter 1

Introduction

This direction contains physical and electrical data necessary for planning and preparing a site. Pre-installation work is defined as site preparation for installation of the GE CT scanner. It is the responsibility of the purchaser to arrange and pay for this work. Pre-installation work includes:

- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of interconnection wiring that is AWG stranded copper. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Ground wires must conform to local codes.
- Any site renovation.
- Alterations and modifications to products not specifically included in the sales contract.

All work must conform to local building and safety codes. Unless specifically mentioned, GE Medical Systems does not provide or install wires, conduits, junction boxes, and ducts as illustrated in this publication.

All CT site plans, preliminary concepts and final working drawings must be reviewed by General Electric Headquarters Architectural Planning prior to construction or approval.

Contact your local General Electric sales representative for complete information regarding your site-specific room layout.

Section 1.0

Site Readiness

The following **MUST** be completed before installation work can begin for a GE CT scanner:

- Finished wall, ceiling, floors & mill work
- Active phone line and network connection
- Power available to A1, w/provision for LOTO



NOTICE An improperly prepared site (i.e., one that is in a state of construction) can result in *increased installation time*.

A CT scanner installed in a dirty environment is more prone to contamination, which can result in *decreased reliability* and *increased scanner downtime*.

Section 2.0

Responsibility of Purchaser

“Pre-installation” is work necessary to plan and prepare a site for installation of equipment. Avoid delay, confusion, and waste of manpower. Complete the following pre-installation work:

- Procure required materials.
- Install required material before delivery of system components.
- Complete all alterations and modifications if not specifically included in the sales contract.
- Complete the Pre-installation Check List in [Chapter 2](#).

Section 3.0

Site Preparation Prior to Equipment Delivery

- Determine room dimensions and verify entry doors are large enough to bring scanner system through.
- Install table/gantry floor duct.
- Install junction boxes of correct size with covers at locations shown in installation plan.
- Install conduit, duct, and raceway.
- Install power supply of correct voltage output and adequate KVA rating. Install local disconnects, including proper over-current protection.
- Install “Unistrut” or other suitable support work for mounting equipment on walls or from ceiling.
- Camera should be on-site at the time of installation.

Chapter 2

Pre-Installation Overview

Before a LightSpeed 4.X system can be installed, all pre-installation requirements must be complete.

Section 1.0

Site Ready for Installation

Site-specific items must be verified before the installation can begin:

1.1 Dust/Dirt Contamination

LightSpeed 4.X systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.

1.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed 4.X components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration's should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

1.3 Walls, Ceiling, and Floor

All walls, ceiling, and flooring must be completed before installation can begin.

1.4 Phone Line

Installation of phone line(s) must be complete and operational.

1.5 Review

LightSpeed 4.X systems use adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads and six (6) auxiliary "leveler" pads located just inside the gantry base, and two each on the front and rear support frame. The auxiliary pads are sometimes referred to as "Jig-feet" and/or "inside levelers". The table has five (5) pads used for leveling it.

Using the GE print to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the template shipped with the system, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the 2 floor templates.
- 2.) Start with the Gantry template—align per the GE print.
- 3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 4.) Check the level of the floor (See [Figure 2-1](#)) across the templates.

Note: Tiles (or other resilient flooring) around all holes will be cut during the installation process.

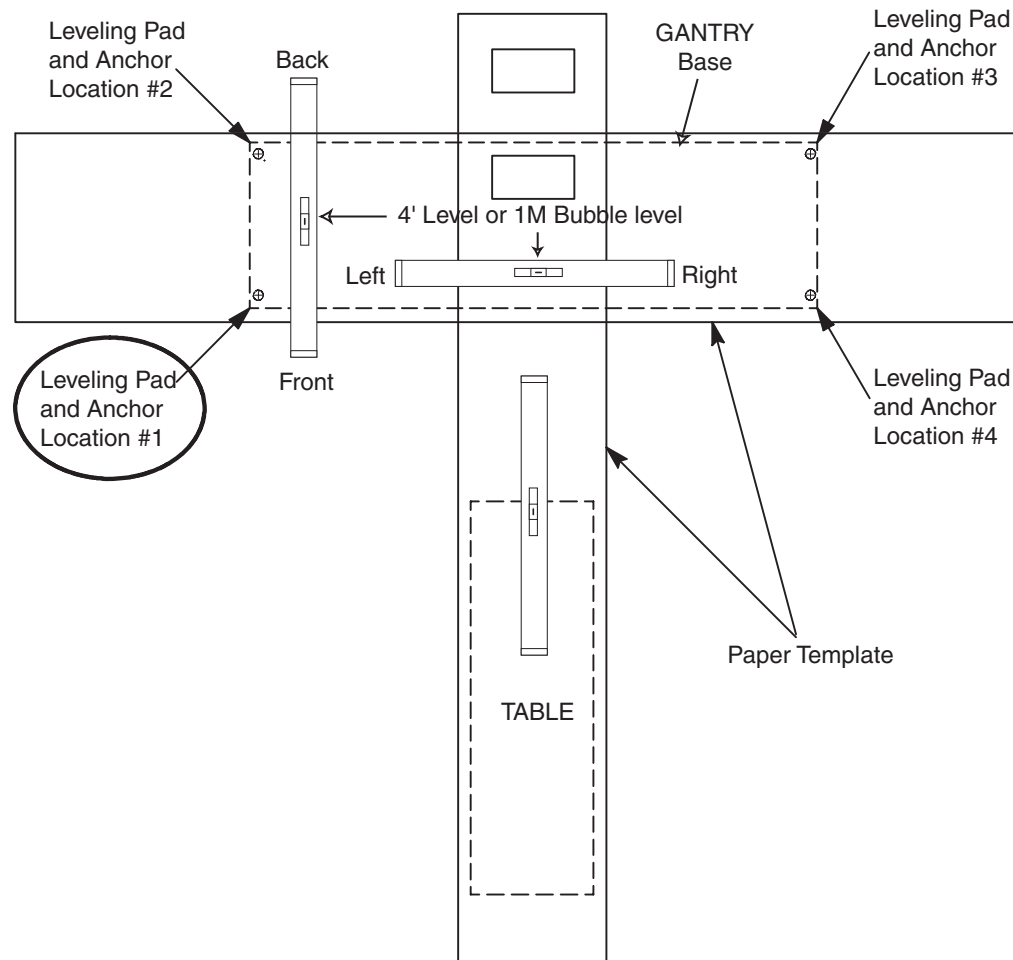


Figure 2-1 Hole Locations

Section 2.0 Pre-Installation Checklist

GE	CUST	
Y N	Y N	DATES
	☐ ☐	Project schedule verified with contractor, facilities department, and GE?
	☐ ☐	Can you meet the committed site ready date?
☐ ☐	☐ ☐	Construction completion date matches delivery date?
☐ ☐	☐ ☐	Power & Ground survey complete, date _____ Hospital contact _____
	☐ ☐	Delivery date is scheduled for: _____
	☐ ☐	First Use date is scheduled for: _____
☐ ☐	☐ ☐	Does the delivery and / or installation date need to be adjusted?
☐ ☐	☐ ☐	Applications dates: On Site scheduled for: _____, Education Center Training: _____

GE	CUST	GENERAL / SITE REQUIREMENTS
Y N	Y N	<i>Must be completed 5 week prior to the delivery</i>
☐ ☐	☐ ☐	Final drawings distributed to the contractors and reviewed with GE?
	☐ ☐	Have any additional requirements or questions of the install been discussed with GE?
	☐ ☐	Final print(s) "signed off" that approve equipment layout / orientation?
	☐ ☐	Person assigned to review and verify that all installation requirements are met?
	☐ ☐	Have the specific site requirements been discussed with the contractor? Refer to the GE final drawings specifications. (See Table 2-1, below)
	☐ ☐	Has the responsibility of cabling, installing, interfacing accessories not on the order been discussed? (<i>Refer to service price pages for GE's support of 3rd party accessories</i>)
	☐ ☐	All 3 rd party vendors identified, notified and scheduled? (i.e., Netcom, 3M, Kodak, Medrad, etc.)
	☐ ☐	Will existing network, modem, and camera cable drops reach new locations / requirements?

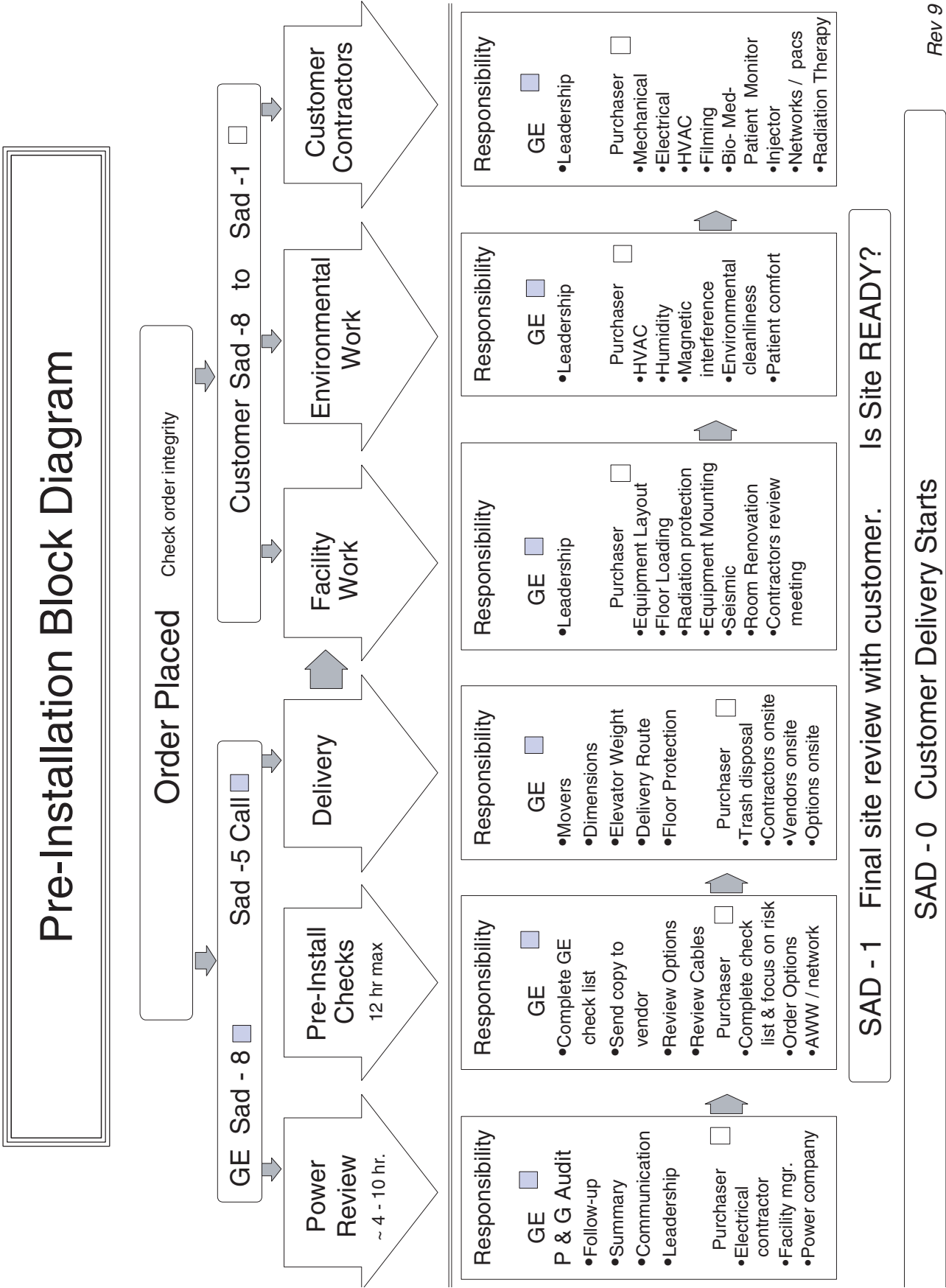
Requirements?	
• Air Conditioning	• Structural
• Electrical	• Other site
All work by contractors must be completed prior to GE delivery	

Table 2-1 Specific Site Requirements

GE		CUST		EQUIPMENT
Y	N	Y	N	
				<i>Must be completed 5 weeks prior to the delivery</i>
☐	☐	☐	☐	Order reviewed for completeness and compatibility with existing equipment.
		☐	☐	Accessories on the order verified for compatibility (i.e. DASMs, cable lengths, laser cameras, AWW work station, injector and other NL items, etc.)
		☐	☐	Interfaces to existing and/or new accessories ordered and planned for accordingly.
		☐	☐	Has the location of any peripherals been determined and planned accordingly with GE representatives? EKG monitor___ Injector control___ Laser camera ___ UPS ___ 2 nd Monitor___
☐	☐	☐	☐	Have any additional services to be provided by GE been discussed and agreed upon?
		☐	☐	Remote monitors ___ AWW relocation ___ Wall mounted accessories ___

GE		CUST		NETWORKING
Y	N	Y	N	
				<i>Must be completed 5 weeks prior to the delivery</i>
☐	☐	☐	☐	Have IP addresses and Host Names been obtained?___ Will a network camera be used?___
☐	☐	☐	☐	CT service telephone line identified and installed for the InSite (<i>Electrical, mechanical, etc.</i>)
☐	☐	☐	☐	Network installed? ___ Network jacks installed and tested?
☐	☐	☐	☐	Network options ordered ___ HIS RIS option ___ DICOM print ___ AWW ___
		☐	☐	Test network connections.

GE		CUST		OTHER
Y	N	Y	N	
				<i>Must be completed prior to the delivery</i>
☐	☐	☐	☐	Arrangements made in the schedule to allow for remodeling, if required. (i.e., wall, floor, or ceiling repair work, painting, other cosmetic finishes)
☐	☐	☐	☐	Have arrangements been made to clean the floor <i>after</i> equipment removal and <i>prior</i> to reinstall?
☐	☐	☐	☐	Is de-installation required? No___ Yes ___ CT removal date _____
☐	☐	☐	☐	Is there a trade-in of existing equipment?
☐	☐	☐	☐	Delivery route identified, and verified with the proper hospital personal? _____
☐	☐	☐	☐	Appropriate arrangements made with traffic for delivery?
☐	☐	☐	☐	Will acceptance testing or Bio-Medical testing be required?
☐	☐	☐	☐	Trash bins available for the removal of papers/ boxes/ etc. during the installation.



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Chapter 3

System Catalog

Section 1.0

Option Catalog Numbers

The following is a list of system options requiring site planning work for the LightSpeed 4.X system. Contact your local GE Medical Systems Sales representative for a complete list of all system options or visit us at <http://www.gemedical.com>. Refer to the instruction manuals supplied with specific options for respective details.

CATALOG NUMBER	OPTION DESCRIPTION
Uninterruptible Power Supplies.	
B7999PS	"Powerware" Un-interruptible Power Supply
B7999PT	"Liebert" Un-interruptible Power Supply
International Dolly Set	
B7850LD	For International customers, if dollies are required.
Rear Cable Cover	
B7850RC	Optional rear cable cover for LightSpeed 4.X upgrades
GE Color Printer	
E7014LA, E8203JA	GE Color Printer includes Cassette
Remote Color Monitors	
B7710WM	LCD monitor.
B7530RC	Remote 20 inch diagonal color monitor.
SmartScore Option	
B7850KC	EKG Monitor and Recording Device
Performance Network Kits	
K9000L	6 Node, 10/100 Mbit Auto Sensing
Switched Network Kit	
B7500PM	ConnectPro Option provides a direct interface to HIS/RIS
Bar Code Reader	
B7540RB	Bar Code Reader
Options and Accessories	
CT Injectors	See Chapter 10, Section 1.0 on page 87 .
Laser Cameras	See Chapter 10, Section 2.0 on page 89
Advantage Windows Workstations	See sales materials
Advantage Windows Options and Accessories	See sales materials
Seismic Kit	
R4390JC	Earthquake zones

Table 3-1 System Options Catalog (Part) Numbers

Section 2.0

Base Scanner System

2.1 Application

The CT scanner system includes hardware and software to support patient data acquisition and image analysis for whole-body computed tomography.

2.2 Configuration

The base scanner system is configured as shown. All scan and analysis functions are controlled from the operator's console (not shown).



Figure 3-1 Base Scanner System

Chapter 4

Room Planning

Section 1.0 Recommended Layouts

Consult your local GE Sales and Service Representative about your specific needs.

Figure 4-1 and Figure 4-2 show two of many possible room layouts (a third example is shown in Figure 5-1, on page 58). For exact equipment sizes, refer to Section 2.0. Remember, sufficient space must be maintained around equipment for full operation, service and safety.

Note:
Other room
arrangements
are possible.

Cable length is an important consideration in room layout. LightSpeed 4.X systems are shipped with standard length cables, with a set of longer cables (B7850LL) available as an option. See the electrical page of the GE print for your specific requirements. (The layout shown in Figure 4-1, below, is for standard length cables.) Note, also, that where possible, the cables should enter the gantry from below.

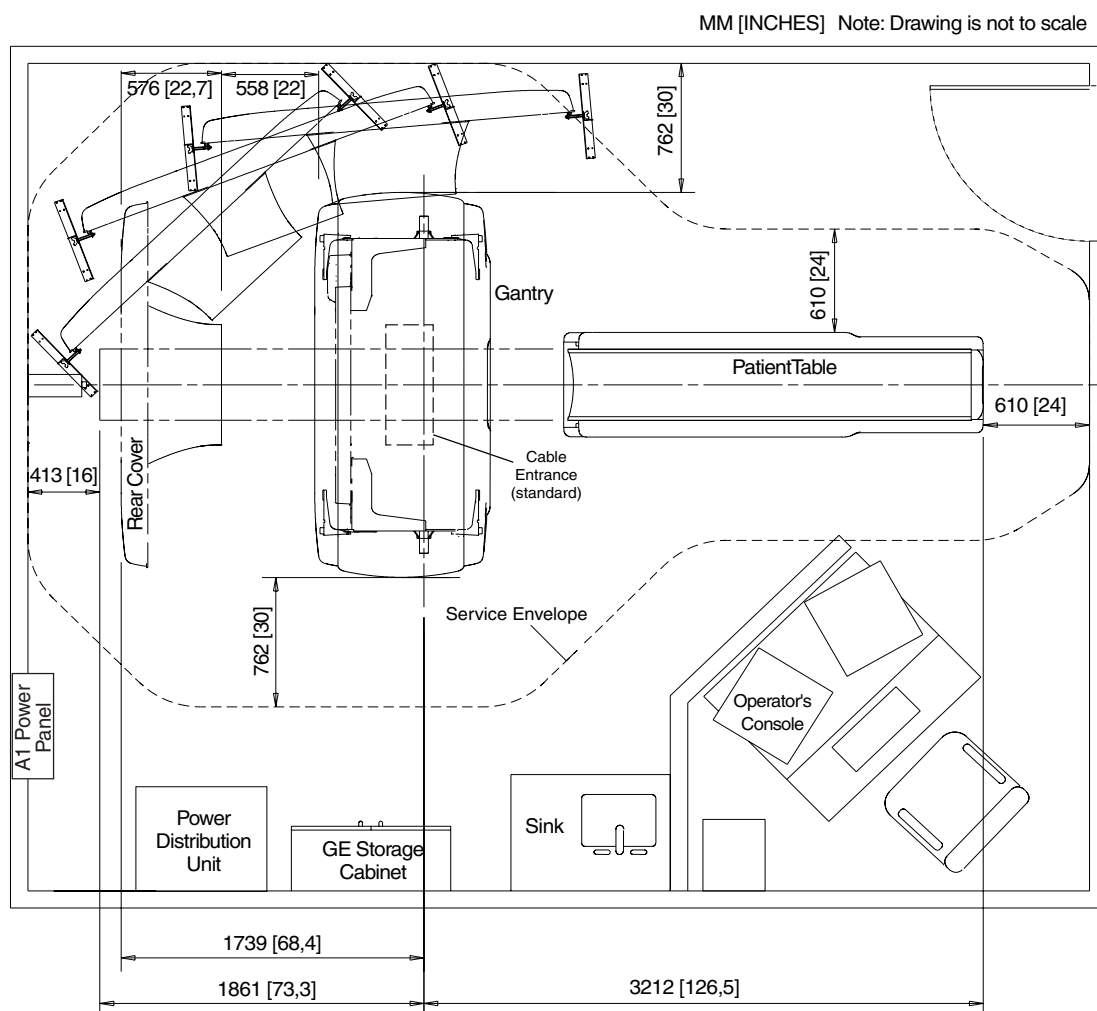


Figure 4-1 Sample Room Layout, Cover Removal Clearances

COMMON DIMENSIONS AND CLEARANCES

System Service

- | | |
|---|-----------------|
| • Back cover removal (rear clearance) | 36.0" (913.5mm) |
| • Front/Back cover removal (side clearance) | 36.0" (913.5mm) |
| • Gantry service side to obstruction | 36.0" (913.5mm) |
| • Area around the PDU | 36.0" (913.5mm) |
| • Area around the A1 breaker box | 42.0" (1067mm) |
| • Area between the table down foot end to obstruction | 36.0" (913.5mm) |

Options

- | | |
|--|------------------|
| Ceiling Pedestal mount lowest point to floor Injector or Monitor | 96.0" (2438.5mm) |
|--|------------------|

System Operation

- | | |
|---|-------------------|
| • Finished ceiling to floor | 108.0" (2743mm) |
| • Table max extension head end with extender from Center Line | 80.0" (2030.0mm) |
| • Table extension head end with extender to obstruction | 6.0" (152.0mm) |
| • Table in lowest position w/cradle at home position to Center Line | 126.5" (3209.0mm) |
| • Back of Console to wall | 6.0" (152.0mm) |
| • Back of PDU to wall | 6.0" (152.0mm) |

Typical Room Dimensions

- | | |
|----------------|-------------------------------|
| • Scan Room | 19.0' X 22.0" (5.80m X 6.60m) |
| • Control Room | 10.0' X 22.0" (3.0m X 6.60m) |

Additional dimensions are available in [Figure 4-3](#) through [Figure 4-7](#) of this document. Consult your local General Electric Installation Specialist for your appropriate room specifications.

1.1 Injector Control

A suitable work area, which is within reach of the operator's console, should be provided for placement of the injector control. Refer to [Figure 4-2](#).

Wall mounted, ceiling mounted and pedestal units need cables to be routed from the gantry area to the console area. The supplied cable is 50 feet in length. Injectors require AC power. Mounts are available in different configurations and lengths. Refer to Injector documentation for detailed installation instructions. Refer to [Figure 4-2](#) for injector placement information.

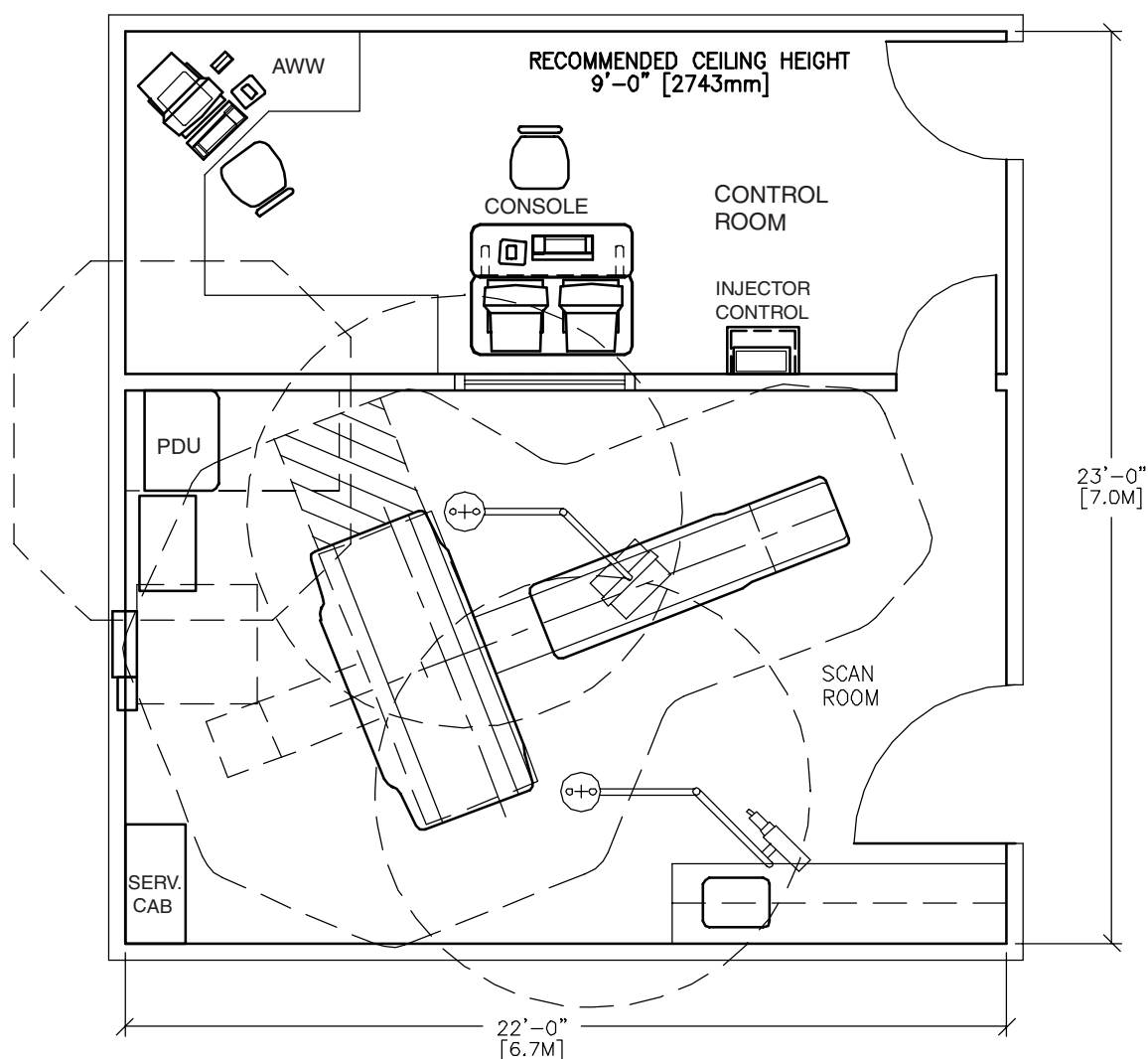


Figure 4-2 Standard Room Layout, with Injector Option

1.2 Storage Cabinet

A storage cabinet is provided by GEMS to store all supplied service equipment (see [Table 4-1](#) for equipment list). This storage cabinet (24" D x 26" W x 42" H) should be located in the scan room suite area, for easy service access. Refer to [Figure 4-1](#) or [Figure 4-2](#).

ITEM	SIZE	WEIGHT (TOTAL)
QA Phantom (water filled)	20 x 15 cm	12 lb
20CM Phantom	20 x 7 cm	13 lb
48CM Phantom	48 x 7 cm	25 lb
Phantom Holder	25 x 25 cm	8 lb
Gantry Display	62 x 13 cm	2 lb
FE Box (Purple)	30 x 38 x 30 cm	15 lb
Rear Cover Dollies	158 x 82 cm	25 lb
Front Cover Dollies	85 x 20 cm and 85 x 15 cm	35 lb
Install Support Kit (box)	30 x 30 x 38 cm	20 lb
Tube Hoist Kit	77 x 8 cm and 38 x 15 cm	20 lb
Gantry Control Pad	27 x 20 cm	4 lb

Table 4-1 Equipment to be stored in storage cabinet

1.3 Advantage Windows Workstation (AWW)

Refer to Pre-Installation Manual (2111833-100) and Installation/Service Manual (2111831-100). Refer to [Figure 4-2](#) for sample layout of AWW option.

1.4 Replacement Component Dimensions

The following are major system components that will be expected to be field replaceable on all *fixed* or *non-fixed* systems.

1.4.1 X-Ray Tube

29" H x 28" W x 48" L 275 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

1.4.2 High Voltage Tank

Single: 23" H x 16" W x 14" L 35 lbs

Dual: 23" H x 27" W x 17" L 70 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front right side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

1.4.3 Detector

17" H x 36" W x 55" L 90 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front right side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

1.4.4 DAS

27" H x 16" W x 27" L 40 lbs

Shipped in a box. Must have unobstructed path to move this box to the front of the gantry and also have working room around the gantry. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

1.4.5 Slip Ring

60" H x 60" W x 10" D 128 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front right side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

1.4.6 Gantry Motor Assembly

17" H x 12" W x 30" D 45 lbs

Shipped in a box. Must have unobstructed path to move this box to the front of the gantry and also have working room around the gantry. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

Section 2.0

Component Dimensions

DESCRIPTION	MODEL/CATALOG NUMBER	WIDTH		DEPTH		HEIGHT	
		INCH	MM	INCH	MM	INCH	MM
System	B7868J*						
Gantry	B7868NG	86.6	2200	39.4	1000	74.6	1895
Table	B7850NT	24.3	617	94	2387		
Power Distribution Unit (PDU)	B7850NP	30	762	22	559	50	1270
Operator's console/ computer	B7868CC or B7868CD (GOC1) B7868GF (GOC2)	48.75	1238	40.15 ¹	1020 ¹	26.78 ²	680 ²
Remote Color Monitor	B7530RC (OPTION)						
Powerware UPS	B7999PS (OPTION)	30	762	16	406	28	711
Advantage Windows	B7997AB (OPTION)						
Workstation		16.5	419	16.1	409	3.1	79
Monitor		19.5	495	19	483	18.6	471
A1		20	508	8	203	31	787
1) Maximum Dimension							
2) Minimum Dimension							

Table 4-2 Dimensions of Components

2.1 Table and Gantry

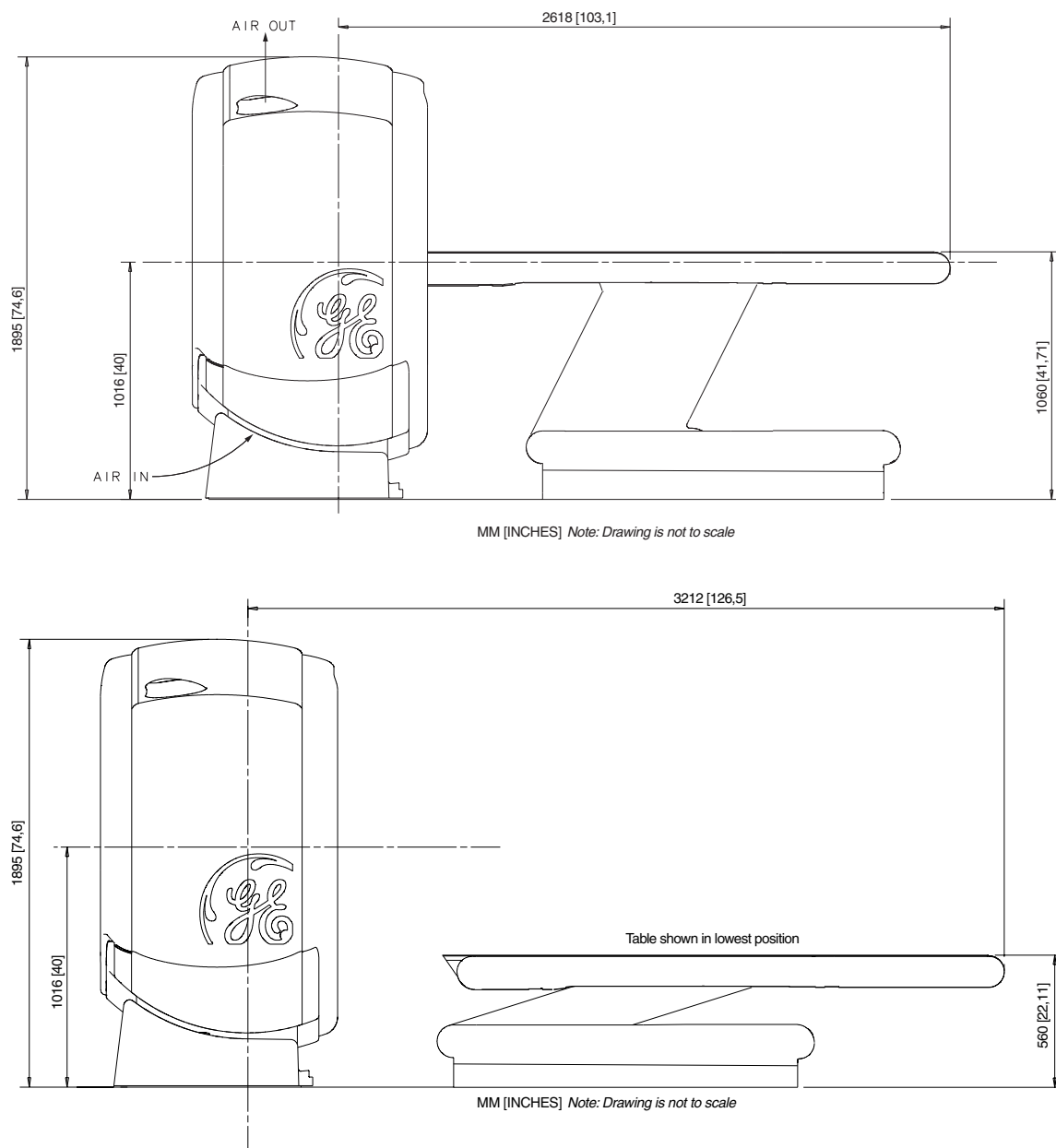


Figure 4-3 Table and Gantry (Side View)

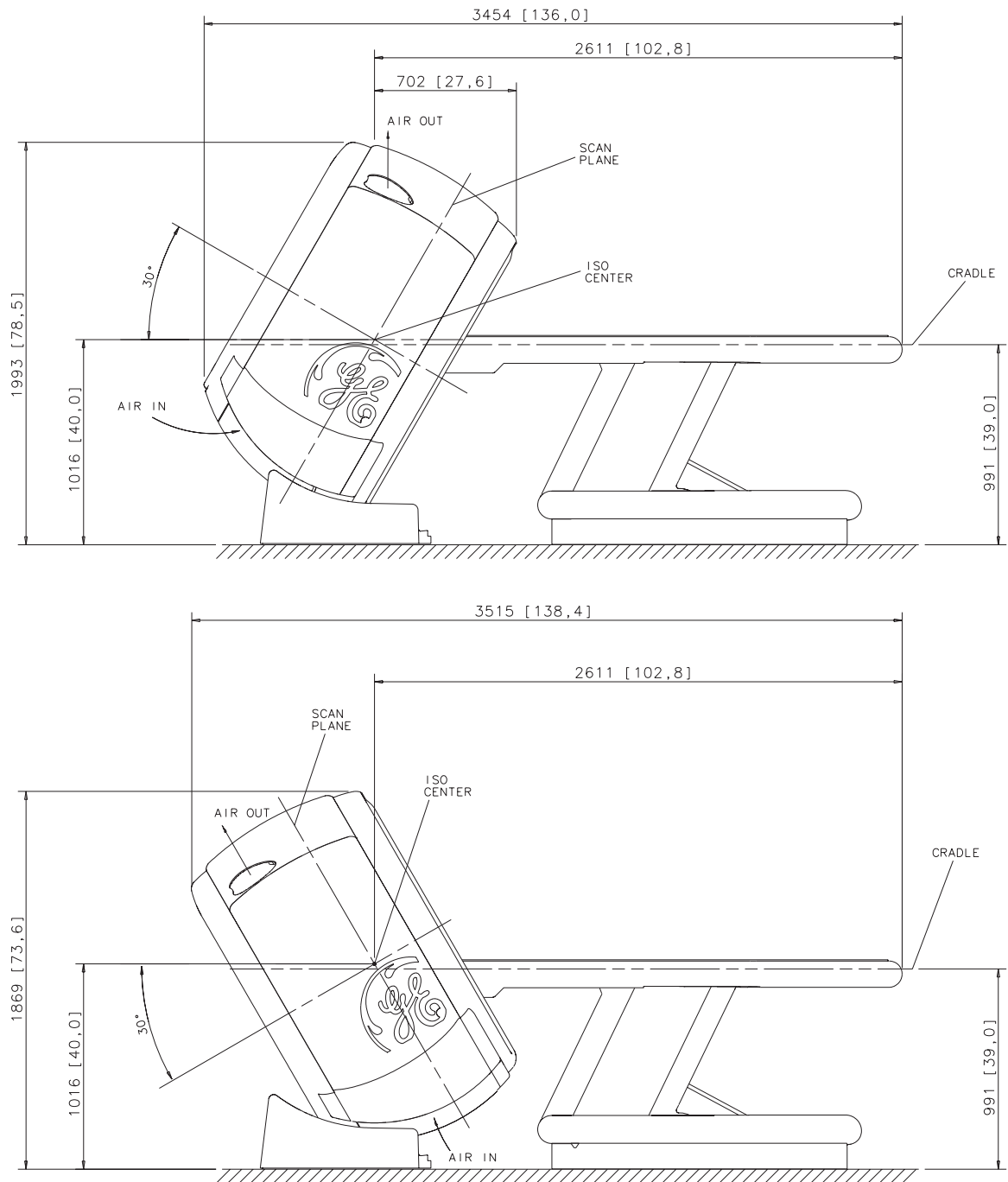


Figure 4-4 Gantry shown tilted +30° (top) and -30° (bottom)

2.2 Power Distribution Unit

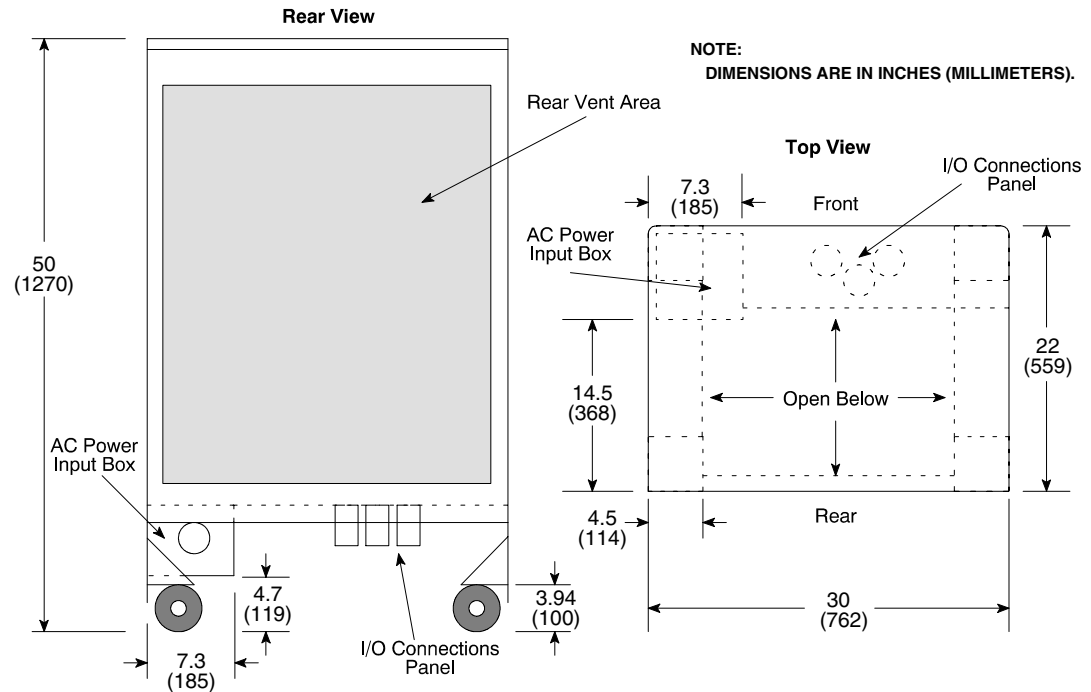


Figure 4-5 Compact Power Distribution Unit (PDU)

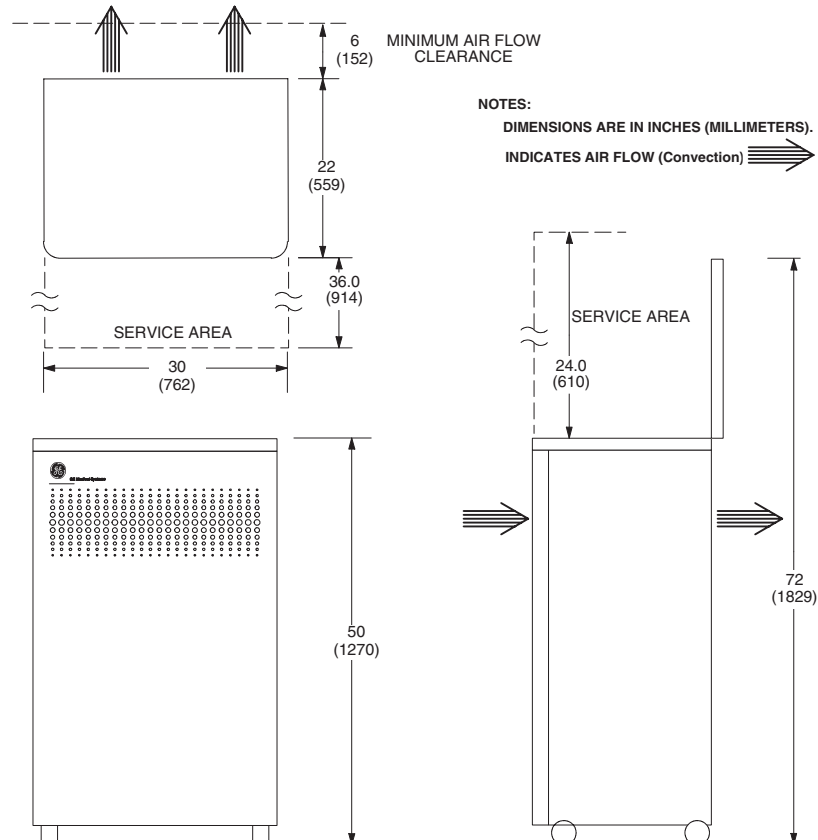


Figure 4-6 Compact Power Distribution Unit (PDU)

2.3 Operator’s Console

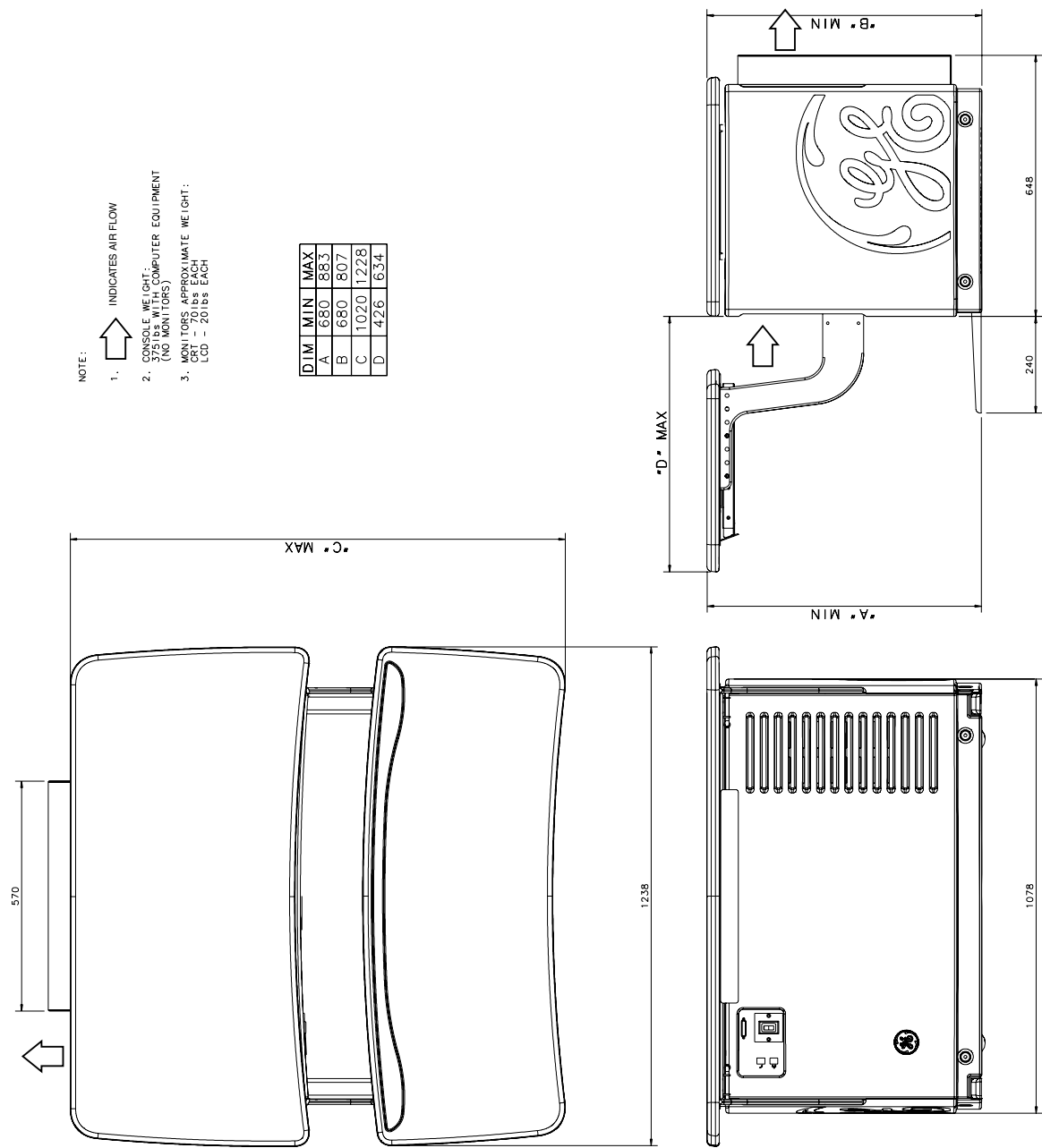


Figure 4-7 Operator’s Console

Section 3.0

Structural Requirements

3.1 Suggested Ceiling Heights

A ceiling height of 108" (2743mm) is recommended. A ceiling height of 92" (2337mm) is the absolute minimum acceptable height for gantry installation.

3.2 Table and Gantry Mounting Requirements



WARNING **POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.



NOTICE **It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements. GE is not responsible for methods other than those listed.**

Table and gantry mounting dimensions are shown in [Figure 6-2](#), [Figure 6-3](#), [Figure 6-4](#). Refer to Chapter 6 for additional details of floor loadings, component weights, and Gantry and Table installation and anchoring.

Anchor gantry and table to floor by a means that will maintain their relative alignment and meet applicable building and other local codes, including seismic structural mounting requirements.

Support areas of the table and gantry must rest on solid concrete or other basic flooring, not resilient tile or carpeting, which will slowly yield over a period of time and disturb alignment of table to gantry.

Factors that could cause misalignment between gantry and table due to floor sag should be considered. The cradle can potentially carry a 450 lb (205 kilogram) patient. Center of gravity changes as cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to Chapter 6 for gantry and table mounting details.

No part of floor surface within table and gantry, nor the two interface areas between table and gantry, should be higher than the support area for table and gantry.

3.3 Floor Anchors

Provided floor anchors are designed for use **ONLY** on concrete floors that meet the 4-inch concrete floor requirements. All other anchoring methods (on floor types other than the 4-inch concrete minimum) must be determined to meet the stated GE minimum load requirements, at the customer's expense, by their structural contractor.

3.4 Floor Strength

Concrete floors must have a minimum strength of $f'c = 2000 \text{ psi}$ ($1.4 \times 10^7 \text{ Pa}$) at 28 days (curing time) for mounting floor anchors. It is the responsibility of each customer to have appropriate tests performed to determine and measure concrete strength.

Note: If installing the GE LS scanner on a floor type thinner than a 4-inch concrete floor, the purchaser, at their expense, shall provide acceptable anchoring and mounting methods that meet all structural specifications provided in sections 3.2 through 3.6 of this Pre-Installation Manual.

3.5 Floor Levelness

The CT Room floor levelness requirement is important for accurate patient positioning. Floor levelness in the Scan Room must not be greater than 0.3125 (8mm) between depression and high spots over any 120 in. (3048 mm) distance within the area of the gantry and the area around the table (see the envelope shown in [Figure 4-1](#) or [Figure 4-2](#)).

3.6 Floor Vibration

The CT equipment may be sensitive to vibration in the frequency range of 0.5 to 20 Hz depending on the amplitude of the vibration. It is the customers responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits. However, it is ultimately the customer/architect/engineer responsibility to design the site solution.

3.6.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s² rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period).

3.6.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s² peak-to-peak.

3.6.3 Equipment Location

To minimize the interference, the CT equipment should be placed on a solid floor, located as far as possible from the following vibration sources:

- Parking lots
- Roadways
- Subways
- Trains
- Hallways
- Elevators
- Heliports
- Hospital power plants containing pumps, motors, air handling equipment and air conditioning units

3.7 Walls: Scan Window

The recommended patient viewing window dimensions are 48 in. wide by 42 in. high (1219 mm x 1067 mm). The location of the window is dependent on the position of Operator Workspace position (see [Figure 4-1](#) and [Figure 4-2](#)).

Note: The operator at the Operator Workspace must be able to view the patient during a scan.

3.8 Mobile Units

It is ultimately the responsibility of the customer/architect/engineer to design the site solution (i.e., levelness, vibration, equipment location and proximity effects of surrounding equipment, such as other mobile, transportable and relocatable units that could contain CT, MR, or other medical devices).

Section 4.0 Network Connections

The CT system is connected to the network through the Console.

- A patch cable (not to exceed 10 feet) should be provided by the customer, and it is used to connect the console to a wall box. (See Notes on [Figure 9-3](#))
- Two phone lines should be provided by the facility. One line is for General Electric service personnel. The second line is to connect to facility for customer use.
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.

Section 5.0 Radiation Protection

Scanner-room shielding requirements should be reviewed by a qualified radiological health physicist taking into consideration equipment placement, weekly projected work-loads, and materials used for construction of walls, floors, ceiling, doors, and windows.

[Figure 4-8](#) depicts measurable radiation levels within the scanning room while scanning a 32 cm CTDI phantom (body) and a 20 cm water phantom (head) with the technique shown.

Note:
Actual
measurements
can vary.

All measurements have an accuracy of $\pm 20\%$ because of measurement equipment, technique, and system-to-system variation.

Use the correction factors shown in [Table 4-3](#) to adjust exposure levels to the usual scan technique at your site.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/100
80 kV	0.21
120 kV	0.71
140 kV	1.0
4 x 3.75mm images	0.82
16 x 0.625 LD 8 x 1.25 LD Fluro 5mm	0.59
4 x 1.25 LD 5mm (1i) Fluro 2.5 mm	0.40
1 x 1.25mm images	0.20
4 x 0.625 LD 1 x 1.25	0.10

Table 4-3 Shielding Requirements Scaling

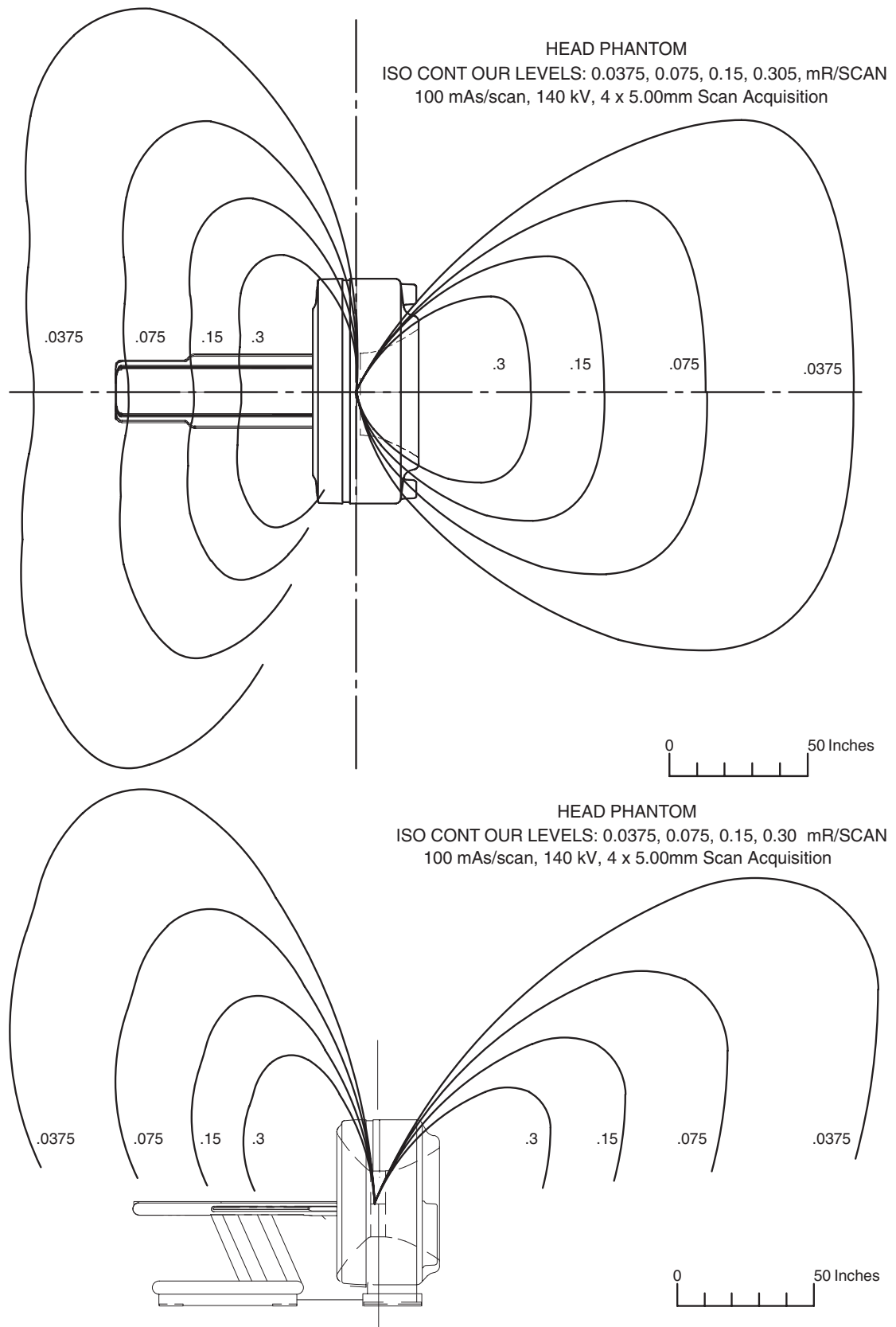


Figure 4-8 Typical Scatter Survey (Head Filter)

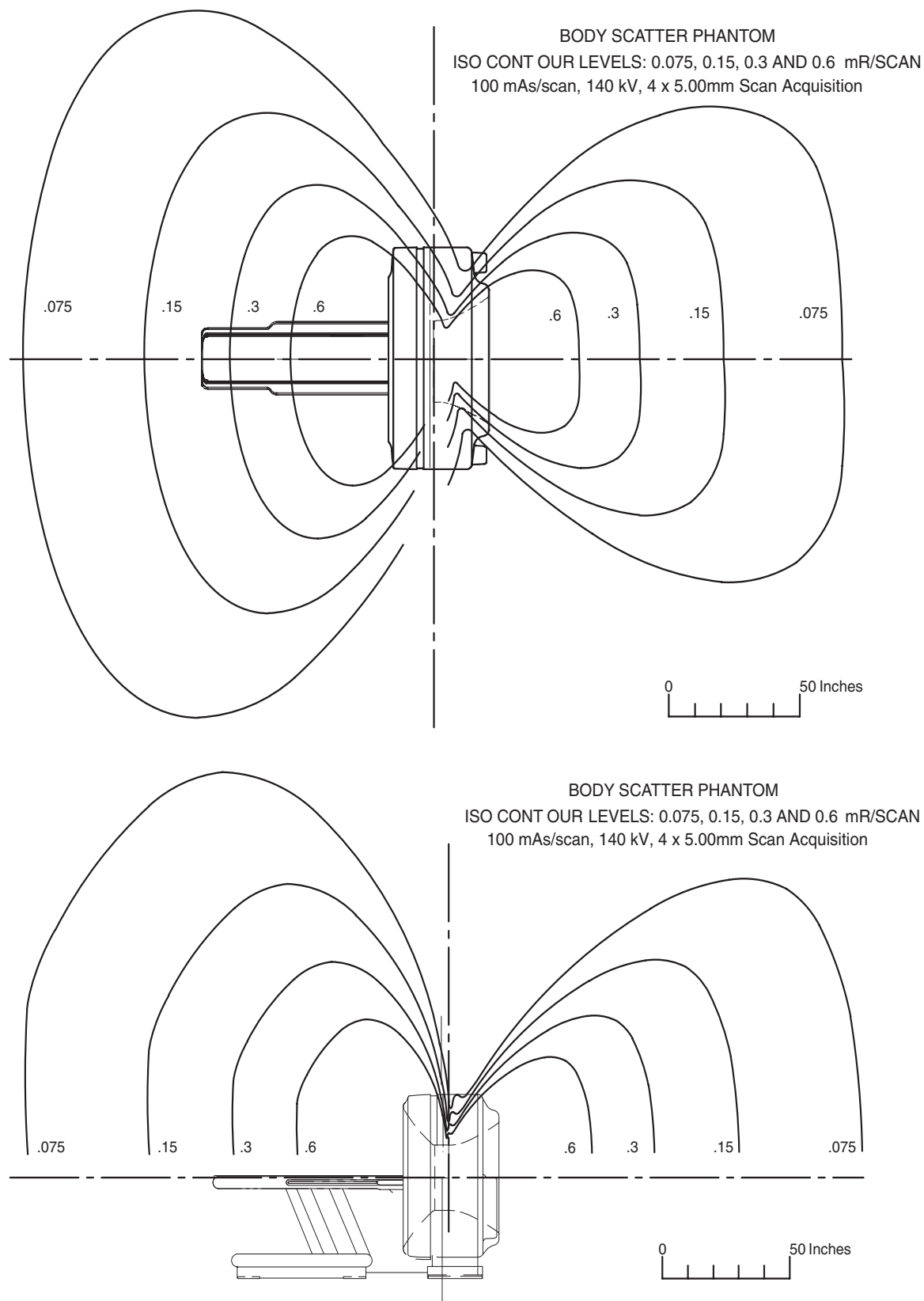


Figure 4-9 Typical Scatter Survey (Body Filter)

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Chapter 5

Environmental Conditions

Ratings and duty cycles of CT subsystems apply if site environment meets the standards of this section. Maintain environmental conditions listed below at all times – including, for example, overnight, weekends and holidays. Shut down the CT system if air conditioning is not working. When system is shut down for major repair, air conditioning may be shut down also.

Section 1.0

Temperature and Humidity Specifications

- Ambient Temperature: (Fahrenheit and Celsius)
 - Scan Room: Maintain a temperature of 70° - 75° F (21° - 26° C), for patient comfort. When scan room is unoccupied, table and gantry temperature limitations are 60° - 75° F (15° - 26° C).
 - Control Room (including Console/Computer): Maintain 60° - 75° F (15° - 26° C).
 - Equipment Room: If a separate equipment room is used to house the PDU, the allowable temperature range is 60° - 84° F (15° - 29° C).
- Store media (cartridges) in long-term storage in same temperature range as host computer.
- Store media in the host computer environment for one-half hour before use.
- Maintain relative humidity of 30%-60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 5° F/hr (3° C/hr).
- The maximum relative humidity rate of change is 5% RH/hr.

Section 2.0

Temperature and Humidity Monitoring

Locate computing subsystems in an area that has the environment specified in [Section 1.0](#), above. First, assess your environment's heat and humidity. If necessary, temporarily install a temperature and humidity recorder close to where the gantry will be installed. Note readings before installation and again after installation to verify true temperature and humidity conditions for your environment. Consider your HVAC needs and redundancy. You may wish to consider an air conditioner with two compressor units rather than one. A backup (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 3.0

Cooling Requirement

Use [Table 5-1](#) to assist in cooling requirements planning. Over half the cooling used by your scanner is required for gantry operation. Locate a wall air-conditioning vent at floor level beside and behind gantry to meet both gantry cooling needs and provide patient comfort. Do not locate any cooling vents directly above the gantry. Air returns above the gantry are recommended.

SYSTEM COMPONENT	ROOM =>	SUITE	
		BTU/HR	WATT
SCANNER SYSTEM			
1. Gantry minimum (See NOTE 1)		22,000	7000
Gantry recommended (See NOTE 1)		25,100	7350
Gantry growth (See NOTE 1)		32,100	9400
2. Table		700	200
3. Power distribution unit		3400	1000
4. Operator's console/computer with two monitors		6800	2000
SYSTEM TOTAL (MINIMUM)		32,900	10,200
SYSTEM TOTAL (RECOMMENDED)		36,000	10,550
SYSTEM TOTAL (GROWTH) (See NOTE 1)		43,000	12,600
SELECTED OPTIONS			
Powerware UPS		2500	732
Liebert UPS		2500	732
Remote Color Monitor		1178	345
Advantage Windows		256	75
ROOM TOTAL (SEE NOTE 2)			
NOTE 1: With 75 scan rotations per patient, minimum gantry cooling is enough for up to three patients per hour. Recommended cooling is enough for up to four patients per hour. Use growth BTU/hr. to size cooling for future, more powerful X-ray tubes.			
NOTE 2: Cooling requirements do not include cooling for room lighting, personnel or non-CT equipment.			

Table 5-1 Cooling Requirements (Worksheet)

Please refer to [Figure 4-3](#), [Figure 4-6](#) and [Figure 4-7](#) in [Chapter 4, Section 2.0](#), for component air flow requirements.

Section 4.0

Altitude

System operating altitude is from mean sea level to 10,000 ft. (3050 meters).

Section 5.0

Electro-Magnetic Interference (EMI)

5.1 Gantry

Locate gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.

5.2 Color Monitor

Locate color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla (500 milligauss) to guarantee color purity and display geometry. See [Figure 5-1](#).

5.3 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity. See [Figure 5-1](#).

5.4 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

5.5 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics (e.g., console or computer equipment - the UPS is not classified as sensitive electronics) within one meter (1m) of the Power Distribution Unit., in any direction (including above or below). See [Figure 5-1](#).

5.6 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Medical Systems Sales & Service for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.

Maintain good screening of cables and cabinets.

5.7 UPS

The Uninterruptable Power Supply (UPS) provides a consistent power supply to various electrical components of the system. Also, it continues to provide electrical power to components during a site-wide power outage so components can be safely shut down. The UPS should be kept at least one meter (1m) away from sensitive electronics (the PDU does not include sensitive electronics).

5.8 Equipment EMI “Envelopes”

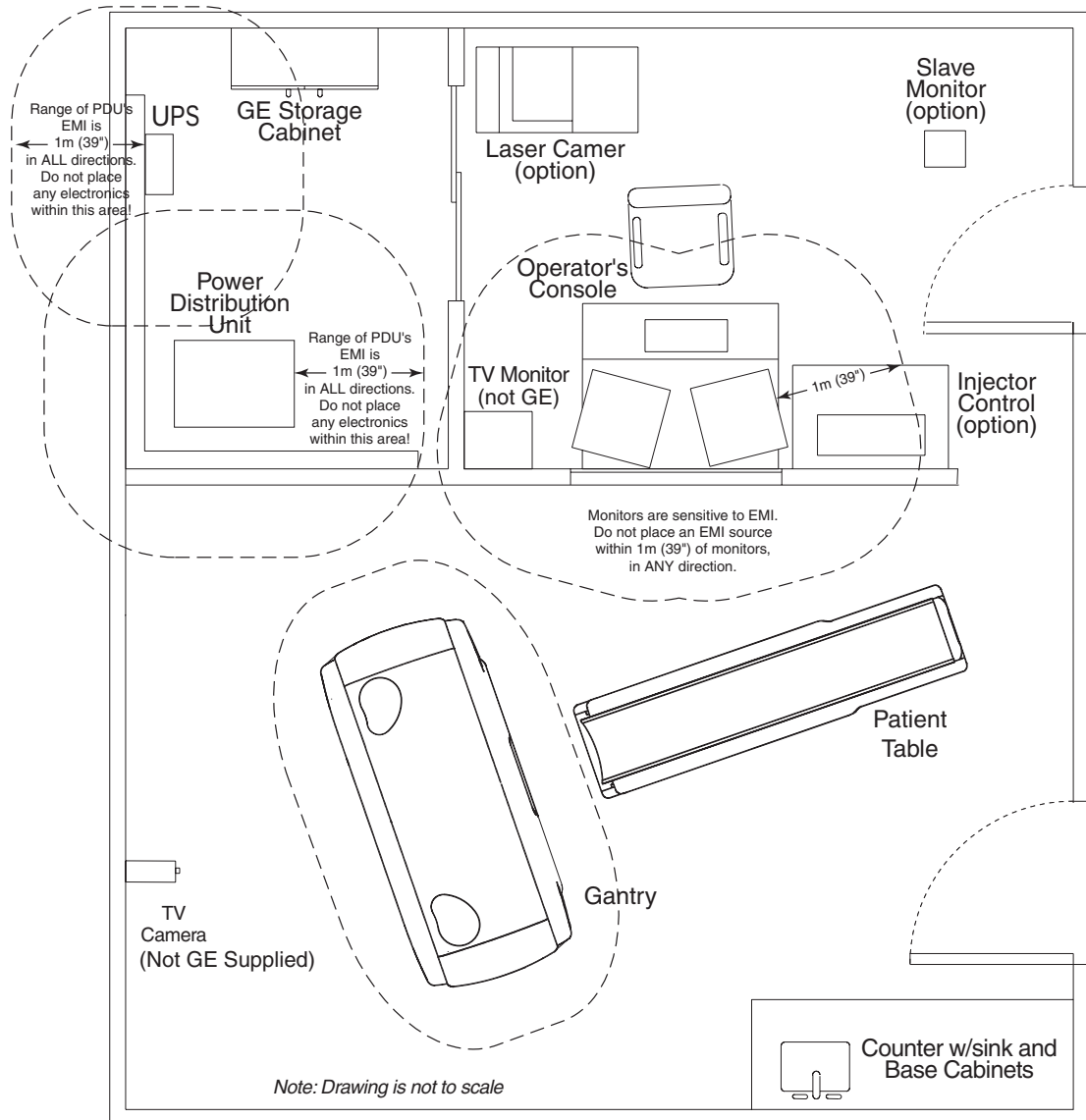


Figure 5-1 Sample Room Layout, showing approximate EMI requirements

Chapter 6

Floor Loading and Weights

Section 1.0 Floor Loading

The LightSpeed 4.X system has a total floor load of approximately 6660 lbs (3021 kg). About 5050 lbs (2291 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly.

[Table 6-1](#) shows CT components with size and weight, floor loading and normal mounting requirements.

ITEM	NET WEIGHT LB (KG)	OVERALL W X D INCH (MM)	WEIGHT/AREA LB/SQ. FT. (KG/M ²)	LOAD PATTERN IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
Gantry	3300 (1500)	86.6 X 39.4 (2200 X 1000)	244 (1188)	Rectangular base plate 24 X 81 (610 X 2057) with four round pads, each 2.5 (63.5) in contact with floor and supporting 1560 lbs (707 kg) center to center.	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at four leveling pads into concrete floor.
Patient Table	1185 (538) Includes 450 (204) Patient	24.3 X 94 (617 X 2387)	175 (852)	Rectangular base 17 X 57.5 (432 X 1460) with five round pads, each 2.5 (63.5) in contact with floor. Worst-case cantilever pull on any bolt is 1235 lb (561 kg).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at three leveling pads into concrete floor.
Power Distribution Unit	740 (336)	30 X 22 (762 X 559)	161 (788)	Four Casters support area of 30 X 22 (762 X 559).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets ² in seismic zones.
Console (w/o monitors)	350 (159)	48 X 39 (1219x991)	58 (281)	Four Casters or Leveling Feet support area of 46 X 19 (1168 X 483).	Casters are for positioning. Set on floor. Four leveling pads may be anchored to floor using angle brackets ² in seismic zones.
Monitor - CRT (ea)	70 (32)				
Monitor - LCD (ea)	20 (9)				

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in [3.2 - Table and Gantry Mounting Requirements](#), on [page 49](#).]
- 2.) Angle brackets are included in the OPTIONAL Seismic Kit (R4390JC) only.

Table 6-1 LightSpeed 4.X System Floor Loading

Section 2.0

Mounting Data, Including Seismic

 **WARNING POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

The following pages show center-of-gravity information for system components:

- Gantry: [Figure 6-2](#)
- Table: [Figure 6-4](#)
- Compact Power Distribution Unit (CPDU): [Figure 6-5](#)
- Operator's Console: [Figure 6-6](#)

Floor mounting hole locations for components that don't have templates are also in this section.

An optional Seismic Kit (R4390JC) is available, for facilities in areas that require one. Customer is responsible for seismic mounting. Refer to all applicable codes in your area.

GE provided floor anchors ([Figure 6-1](#)) are designed to be used **ONLY** on concrete floors that meet the concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 3 inches at each anchor point. Any anchors having more than 1 inch of thread showing above the nut, when torqued to 55 ft.-lbs, shall have a second anchor installed in the closest adjacent hole. the second anchor shall be installed to the standard depth and torque specifications.

MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum Floor Thickness:	4 inches	4 inches
Recommended Drilling Depth:	3-¾ inches	3-¾ inches
Average Anchor Embedment:	3-½ inches	3-½ inches
Minimum Anchor Embedment:	3 inches	3 inches
Available Alternate Anchor Locations:	Yes	Yes
Shipped Anchor Size:	8 inches	8 inches
Alternate Anchoring Methods:	Yes (see notes, below)	Yes (see notes, below)
Floor Levelness Requirement:	0.3125 (8mm) over 10 feet	0.3125 (8mm) over 10 feet

Table 6-2 Gantry and Table Mounting Requirements

- Note: If the Installers cannot set at least three anchors for the table, the installer must inform the customer that the minimum anchoring cannot be met, and a structural engineering contractor is required to determine the anchoring method and certify that their anchoring meets the stated GE minimum load requirement and torque specifications.
- Note: All other anchoring methods on floor types other than the concrete minimum must be determined at the customer's expense by a structural engineering contractor, and anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.
- Note: If installing the GE LS scanner on a floor type other than a 4 inch concrete floor, all structural specifications in this document must be reviewed and met.

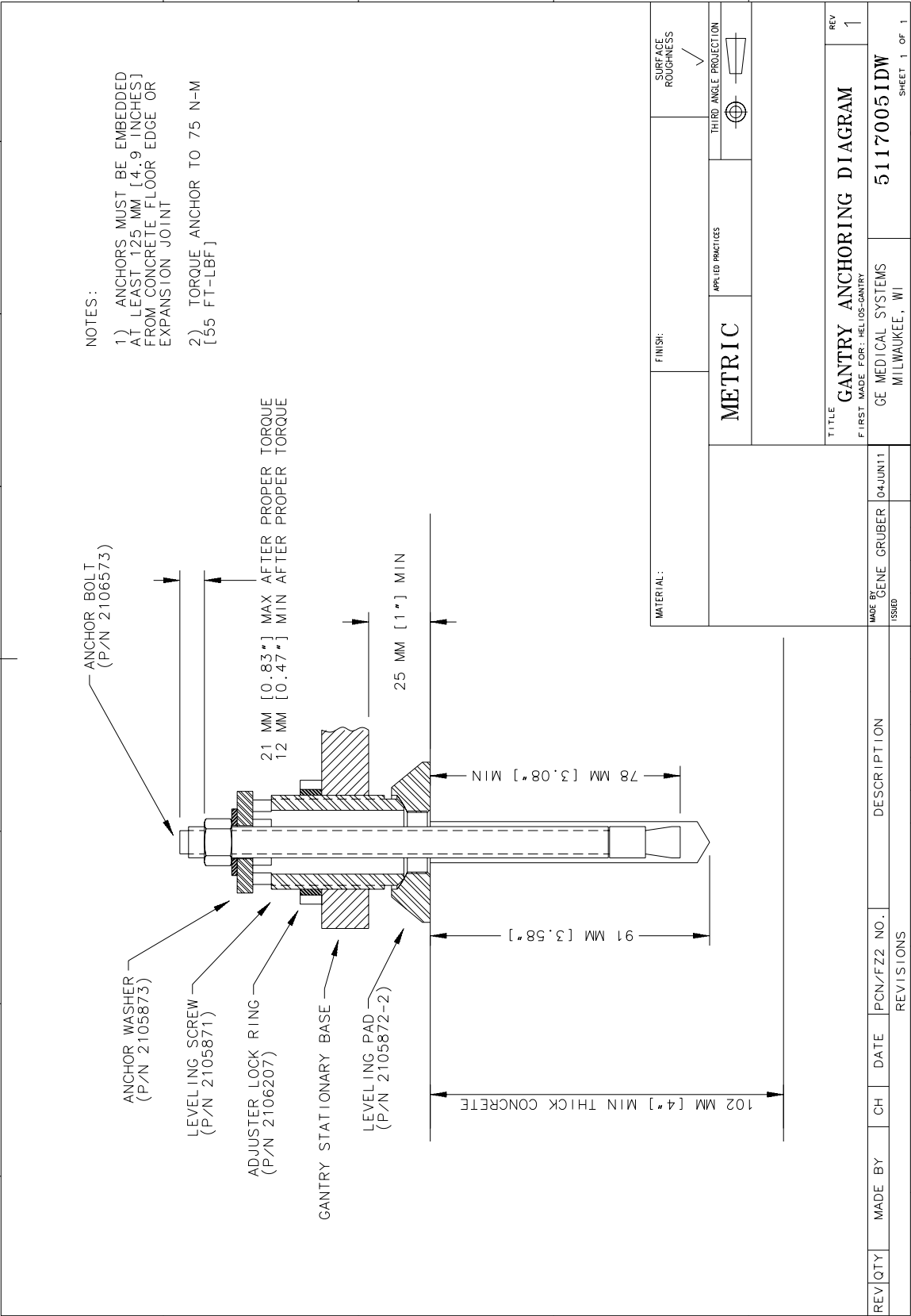


Figure 6-1 Typical Floor Anchor, Gantry and Table

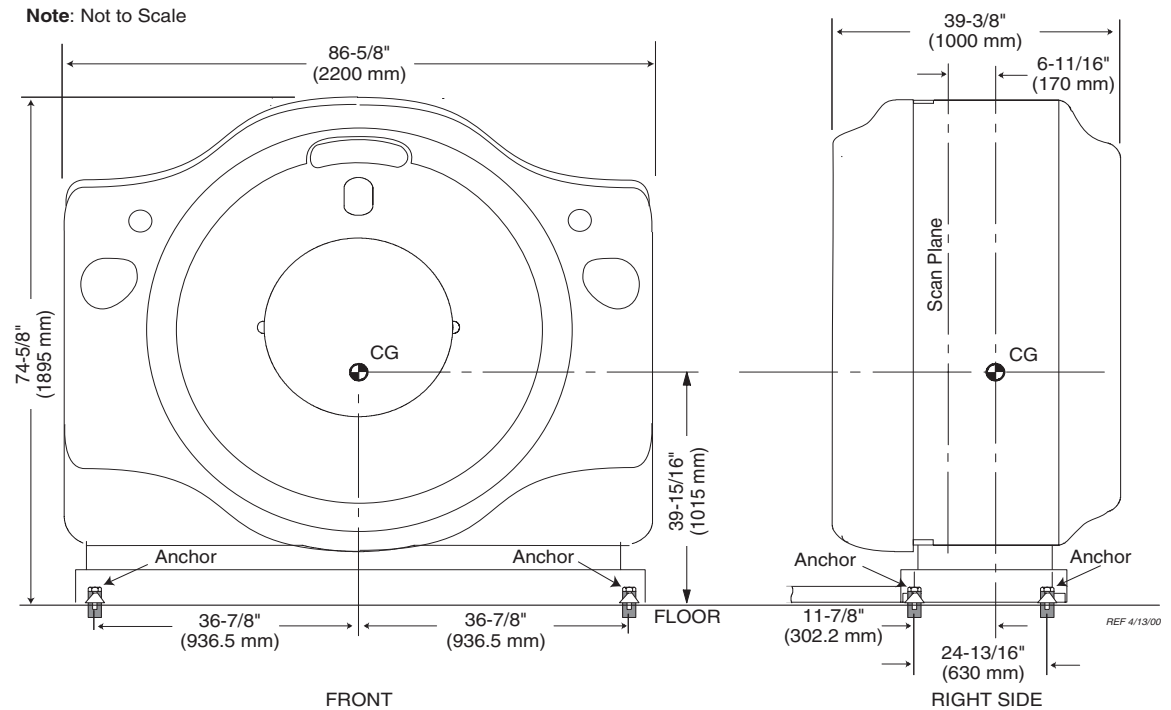
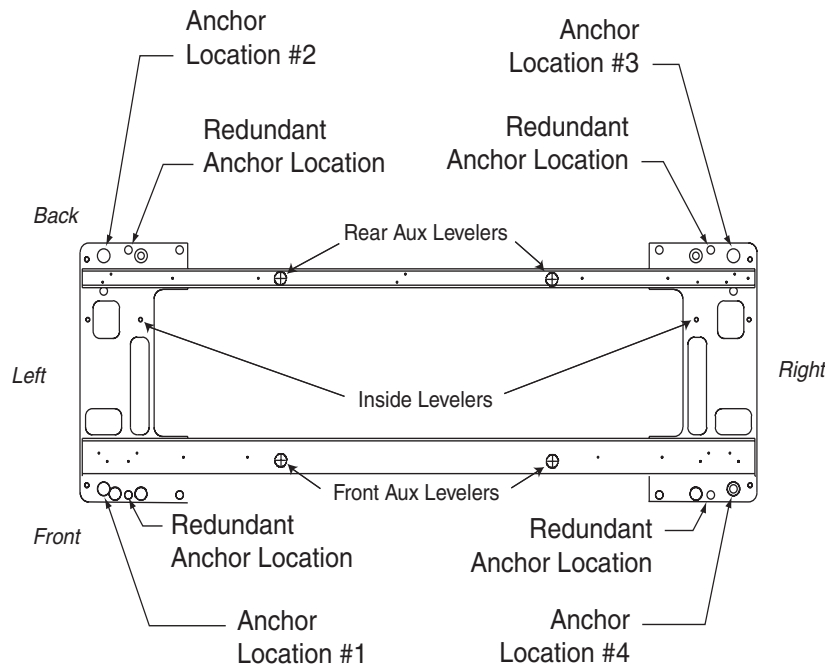


Figure 6-2 Gantry (CT2)



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm).
Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 6-3 Gantry Leveler and Adjuster Locations

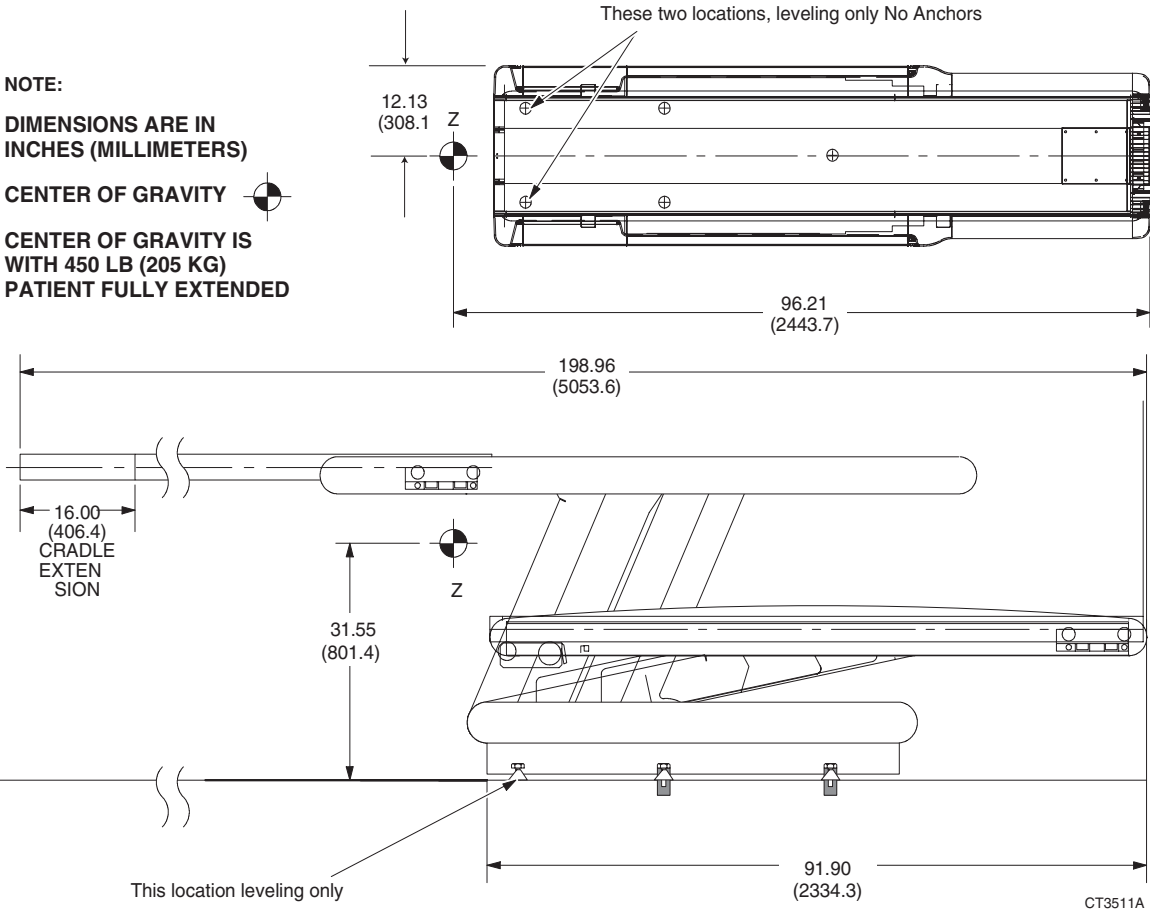
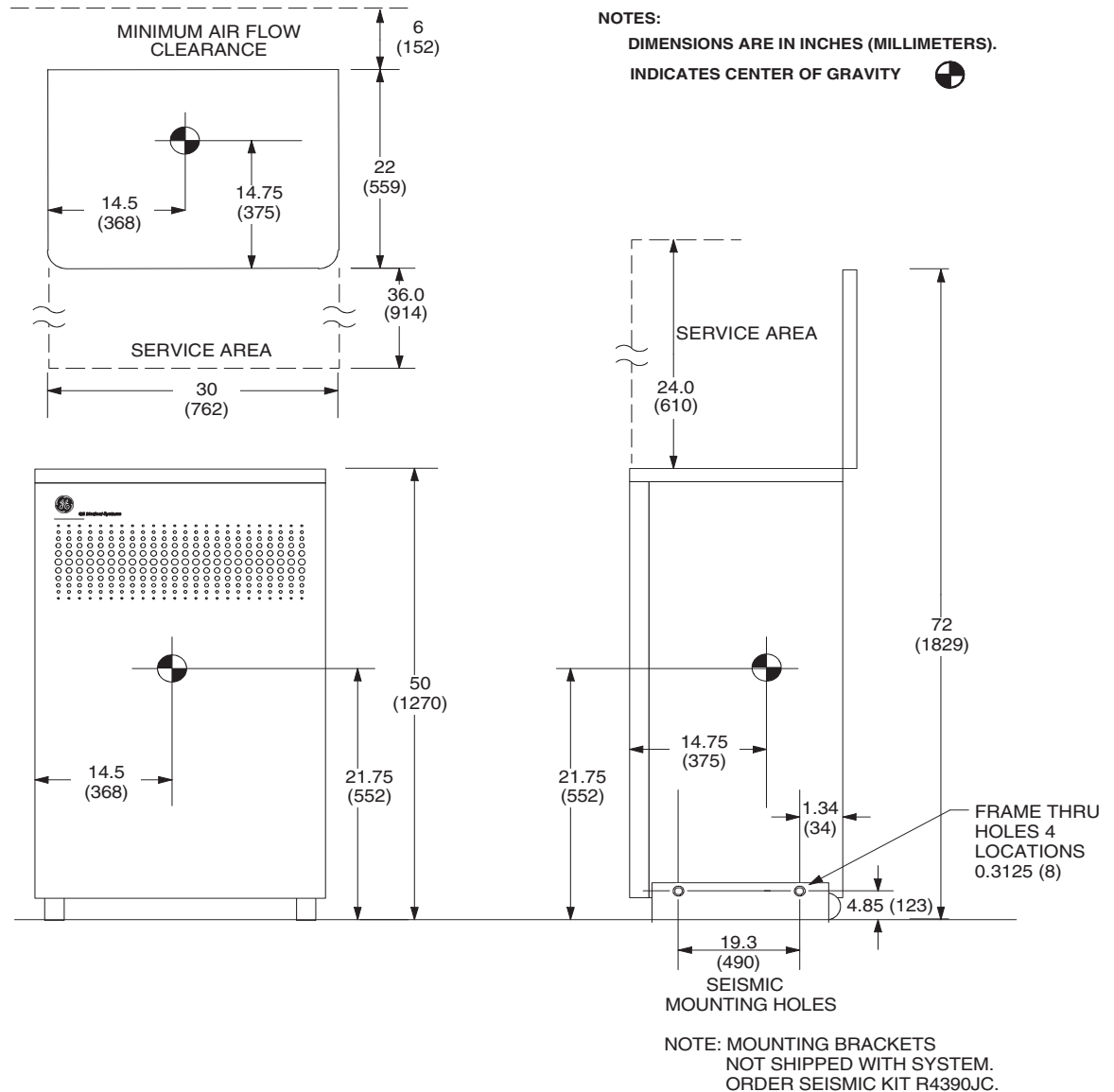


Figure 6-4 Patient Table (CT1)



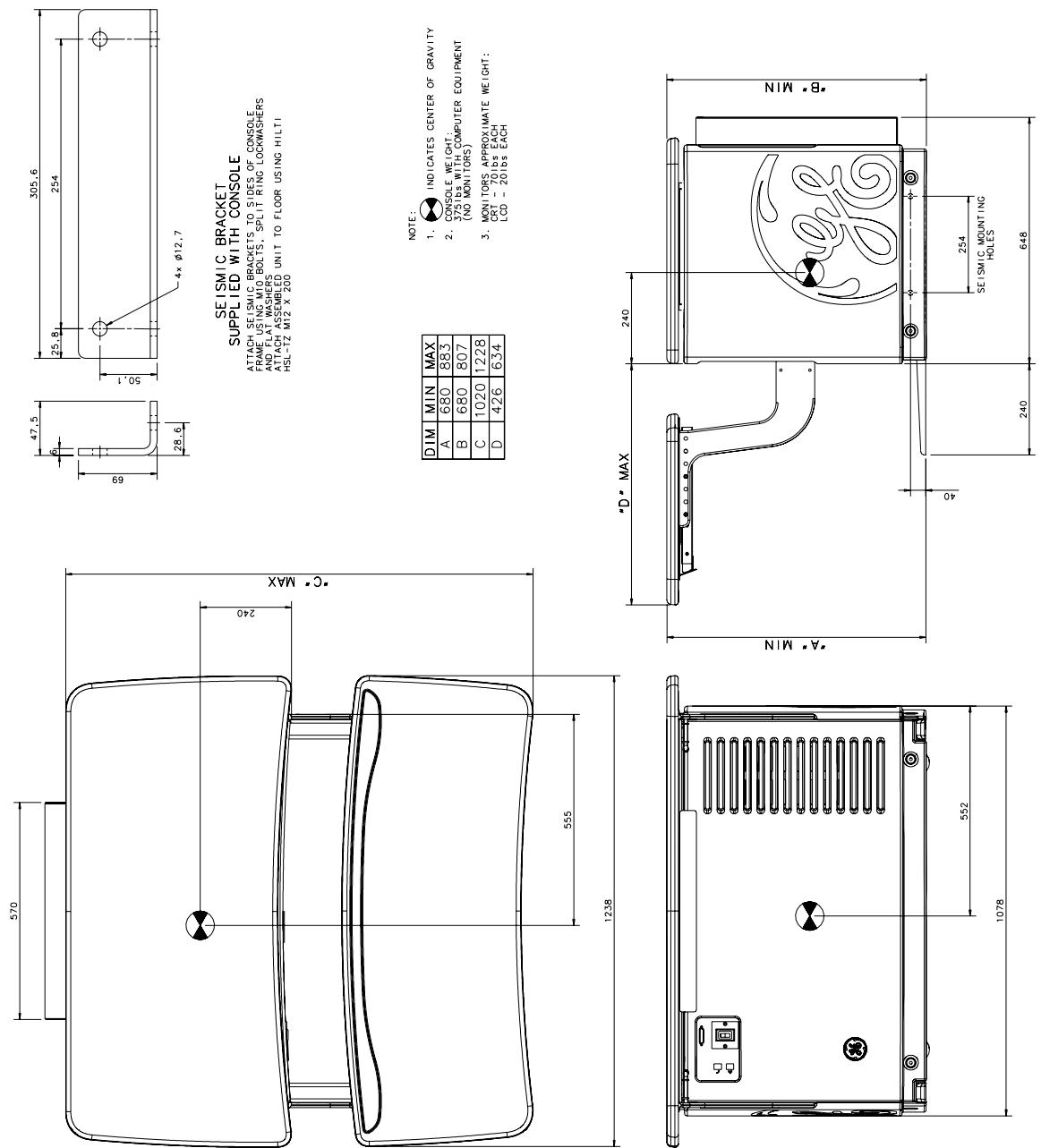


Figure 6-6 Operator's Console



Chapter 7

Delivery Data



WARNING SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0 Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 20 shipping containers which include dollies, skids and boxes without skids.

Section 2.0 Delivery/Shipping Requirements & Considerations

The LightSpeed 1.3 system is not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping or hoisting.

The Gantry, Console, Table and PDU must NEVER be dropped. A drop from a height greater than one half inch ($\frac{1}{2}$ ") may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the Gantry, Dock to Dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, the system may be moved via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces.

When moving Gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the Gantry at the slowest reasonable rate. See [Figure 7-1](#).

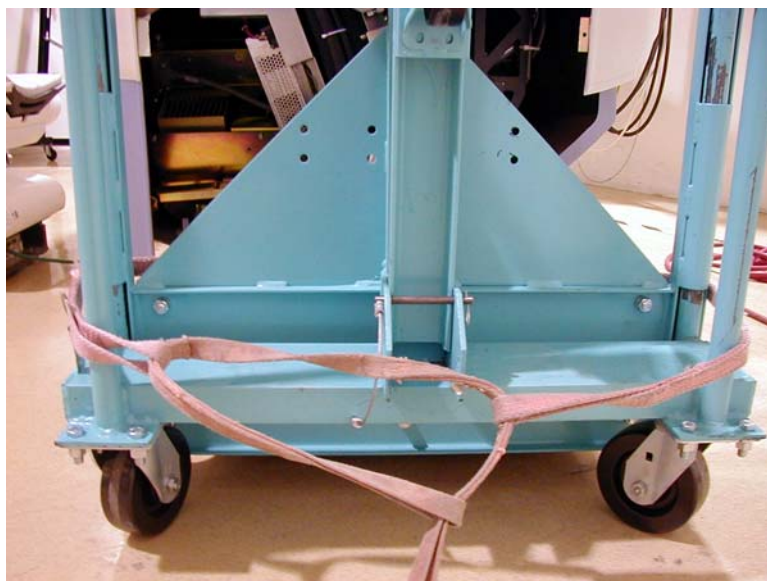


Figure 7-1 Gantry Strap Location

The LightSpeed 1.3 System—including Gantry, Console, Table and PDU—is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the Console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive or octane failure) may not be evident until after system installation is complete.

All system components must remain upright at all times, and must not be tipped. Nor should the Gantry be hoisted. The Gantry should be moved by rolling it on its dollies only. Movement through hallways, doorways, elevators, etc., must be done without tipping or lifting the Gantry.

Protection for flooring along the move path (from dock to scan room) is advised.

Section 3.0

Site Environmental Considerations

3.1 Dust/Dirt Contamination

LightSpeed 1.3 systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.



NOTICE The act of installing a GE CT scanner in a construction (i.e., unfinished) site will likely result in the following adverse effects:

- Increased installation time
- Decreased installation quality
- Increased scanner downtime, due to increased service calls

3.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed 1.3 CT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer’s requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration’s should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0 For Crated Delivery

The standard CT system packing configuration for air shipment is 6 crates, as listed in [Table 7-1](#), below. Note that specific shipping configurations may differ, depending on options ordered. Total weight of the basic system, including pre-installation kit, is 8,182 lbs (3713 kg). It is shipped in wooden crating.

CRATE (#)	HEIGHT IN (CM)	WIDTH IN (CM)	DEPTH IN (CM)	WEIGHT LB (KG)
Gantry (1)	88 (224)	57 (145)	97 (246)	3500* (1588*)
Table (2)	81 (206)	38 (97)	103 (262)	1481 (672)
PDU (3)	60 (158)	37 (94)	40 (102)	800 (363)
Console (4)	51 (130)	57 (145)	62 (157)	992 (450)
Cables/ Collector (5)	65 (165)	51 (130)	44 (112)	634 (288)
Monitors & Options (6)	65 (165)	51 (130)	44 (112)	775 (352)

* If dollies are ordered (and shipped), add 500 lb (227 kg).

Table 7-1 Typical Crated Delivery Sizes and Weights

Section 5.0 Storage Requirements

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust. Storage temperature should not exceed 40° to +80° F (-4° to +27° C). Maintain relative humidity (non-condensing) between 20% and 60%.

Do not store a LightSpeed 1.3 system in a non-controlled environment for extended periods of time. Meeting this requirement prevents rust and corrosion from forming on bearing surfaces due to condensation.

Equipment kept in a moving van while awaiting delivery is considered to be in storage. All temperature and humidity restrictions (including “Extreme Temperature Storage” defined below) apply to equipment in such situations.

EXTREME TEMPERATURE STORAGE

Extreme storage temperatures are defined as below 32° F or above 110° F, without humidity control. CT systems must not be exposed to extreme temperatures while in storage.

Section 6.0

System Transportation

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 20% to 80%

Section 7.0

Extreme Temperature Transportation and Delivery

7.1 Delivery Considerations



NOTICE Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.

- 1.) Avoid all sub-zero and extreme heat van delivery lay-overs. Avoid deliveries where van temperatures reach below zero degrees or above 130 degrees.
- 2.) Special handling measures should be taken to prevent equipment damage, for all extreme temperature deliveries.
- 3.) Use special care to avoid delivery routes that may cause excessive shock or vibration. Choose routes that will prevent dropping or tipping system components.
- 4.) Deliver the CT console to the scan suite on the protective shipping skid, to prevent damage.

7.2 Notes to Mechanical Installers and Field Engineers

- 1.) Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to unpacking of equipment, including all boxes and accessories.
- 2.) Allow an additional 12 or more hours to mechanically install system with no power applied.
- 3.) If moisture is present on any component after the mechanical installation is complete, allow additional time for all moisture to evaporate before system power up.
- 4.) All the detector temperature to stabilize with power ON, until normal operating temperature is reached, as displayed on the home page GUI.

Section 8.0

Gantry Considerations

The gantry is shipped with most covers installed. The assembly is mounted between two dollies. See . Two side rails are bolted to the dollies to stabilize dollies and protect gantry. Use dolly elevating casters to lift gantry off its base and roll it into position.



Figure 7-2 Gantry with Shipping Dollies and Side Rails

Door Openings. Clear door openings for moving equipment into building must be 42 X 82 in. (1067 X 2083 mm) minimum, if there is an 8 ft. (2439 mm) corridor width.

Elevator requirements. To move gantry from receiving location to scanning room, consider elevator capacity and size. By removing side rails and one dolly after gantry is placed in elevator, gantry width/length and elevator depth requirements are reduced. Contact a representative of elevator manufacturer if gantry weight exceeds elevator capacity.

CONFIGURATION	LENGTH IN (CM)	WIDTH IN (CM)	HEIGHT IN (CM)
Dollies On, Side Rails On	114 (290)	51 (129)	79 (200)
Dollies On, Side Rails Removed	114 (290)	39.4 (100)	79 (200)

Table 7-2 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door size for gantry with covers and dollies attached, but side rails removed, is 40 inches (101.6 cm).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Dollies: Typically, dollies are used on the gantry, table, and console for domestic shipments. Once the gantry, table and console are installed at the site, return dollies to GE using the shipping document located in Box #1.

Dollies can be purchased for international shipments (B7850LD) to be used at the customer site. After the system has been removed from the crates, dollies shipped with international shipments **only** are not to be shipped back to GE in Milwaukee, WI, USA, but to be retained to the local GE office or warehouse.

Section 9.0 Table Considerations

The table is shipped without side covers installed. Covers are shipped in four separate boxes. The table is mounted between two dollies.

Table shipping dimensions are 101" (2565 mm) long, 32" (81 mm) wide, and 47" (119 mm) high.

Section 10.0 Console Considerations

Although not installed, the keyboard table is shipped with the console.

Console shipping dimensions are 40" (1016 mm) deep, 53" (1346 mm) wide and 41" (1041 mm) high.



NOTICE Do not lift the console by the monitor table top to remove from the shipping skid.



Figure 7-3 Console ready to be unloaded from shipping skid.

Chapter 8

Power Requirements

Section 1.0 Introduction

The power distribution unit (PDU) supplied with LightSpeed 4.X systems transforms and distributes power to all system components. The PDU is the only power entry point required to operate the system.

Power wiring between the facility main distribution panel and the PDU should be kept as short as possible. This minimizes voltage regulation effects.

Carefully consider advantages and disadvantages of conduits, floor ducts and surface raceways for running cables. Make cable passageways large enough to install any cable with all other cables already installed.

When routing power cables, all three phase wires and ground must be run in the same conduit or raceway duct-work. Power cables should be routed separately from system control cables (for example, use a separate trough in duct).

Section 2.0 System Input Power

2.1 Facility Source

Power to the system should be supplied by a dedicated feeder from the nearest Main Distribution Panel (MDP). A protective disconnect device must be provided in the power line supplying the PDU in accordance with National Electric Code and applicable local codes.

Note:
Lockout/tagout
provision
required

The disconnect device must be located within forty feet of the PDU, visible to PDU service personnel, and must have provision for tagout / lockout. It is identified as “A1” in the interconnection schematic diagrams. The GE part number for the disconnect is R4502AD.

The rating of the disconnect device depends on the nominal line voltage. It must provide over-current protection and have a low voltage release, with multi-point remote control capability. Refer to [Section 3.0: “Recommended Power Distribution System”](#), for minimum rated capacity requirements and suggested device.

2.2 Main Disconnect Control

Customer supplied emergency off buttons are to be mounted in the CT Room near each exit 60 in. (1424 mm) from the floor and connected to the protective disconnect device in order to disable the power to all CT system equipment in emergency situations. This button is to be clearly labeled “Emergency Off” and visible to personnel in the CT Room. It is important that the button be labeled “off” and not “stop” since there exists an “Emergency Stop” button in the LightSpeed 4.X system that disables output power to the CT system equipment in the patient area of the CT system. An additional emergency button should be mounted in the computer/equipment room near the exit door.

Customer supplied main disconnect controller (MDC) must be lockable to meet OSHA requirements for power Lockout/Tagout requirements. An optional MDC is available from GE providing for the disconnect of the facility power to the PDU, including the emergency buttons, and contacts for an interlock to the air-conditioning units in the computer/equipment room.

The Main Disconnect Control Panel must be UL or other nationally recognizes testing organization listed and labeled in accordance with the latest National Electrical Code (NEC) specifications.

2.3 Configuration

LightSpeed 4.X systems are designed to operate on three-phase, **four-wire** wye power. A ground referenced wye source produces the lowest leakage currents and is preferred. However, the neutral wire does not need to be run to the system, i.e., four-wire connection. (A dummy terminal is provided for ‘parking’ the neutral wire in the event a five-wire service is already installed at the site.)

2.4 Rating

Voltage	380 to 480 VAC (see note, below)
Capacity	90 KVA @ 85% P.F.
Frequency	50 or 60 Hz +/- 1.0 Hz

Note: The absolute range of line voltage at the input to the PDU must remain within one of the ranges shown in [Table 8-1](#) at all times.

2.5 Regulation

The size of the facility transformer and feeder wires determine load regulation presented to the system. Total load regulation as measured at the PDU input terminals must not exceed 6%.

2.6 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.7 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in [Table 8-1](#). Limit maximum transient voltage to 1500V peak.

2.8 Microcuts

LightSpeed 4.X systems are generally unaffected by microcuts.

2.9 Grounding

The ground to the CT system shall originate at the system power source, (i.e., transformer or first access point of power into a facility) and be continuous to the CT system power disconnect in the room. **A dedicated 1/0 (55mm²) or larger insulated copper ground wire must be run with the phase wires from the main distribution panel to the PDU.** These grounds can be spliced with “High Compression Fittings” and should be terminated at each distribution panel it passes through. When the ground is broken for a connection to a panel, it shall be connected into an approved non-insulated grounding block with the incoming and outgoing ground in this same grounding block, which is then connected to the steel panel, never using the steel or other material of the panel as the block.

The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and **earth must not exceed 2 ohms.** Resistance between any two grounded devices must not exceed 0.1 Ohm to ensure equal potential ground system within the CT Room.

Section 3.0

Recommended Power Distribution System

A single-unit installation, where the distribution transformer and feeder are dedicated to the LightSpeed 4.X system is recommended. In this case, the minimum recommended transformer size is 112.5 KVA, rated 3.2% regulation at unity power factor. The minimum recommended feeder size and over-current protection device ampacity based on line voltage is shown in [Table 8-1](#).

- Maximum power demand = 90 kVA @ 0.85 PF at a selected technic of 140 kV, 380 mA.
- Average (continuous) power demand at maximum duty cycle = 20 kVa.
- Maximum allowable total source regulation = 6%.

CATALOG >	B7850NP (LIGHTSPEED SERIES PDU)					
Nominal Line Voltage (VAC)	380	400	420	440	460	480
Voltage Range (VAC) +/- 8%	350-410	368-432	386-454	405-475	423-497	442-518
Average Line Current (A)	30	29	27	26	25	24
Momentary Line Current (A) @ Nominal Line Voltage	137	130	124	118	113	108
Maximum Line Current (A) @ Low Line Voltage	149	141	135	128	123	117
Primary Disconnect Device (A1) Over-current Protection (A) ⁶	110	110	100	100	90	90
Feeder Length (MDP to A1) ¹						
0-100 ft (0-30.5 m)	2 (35) ³	2 (35) ³	3 (30) ³	3 (30) ³	3 (30) ³	3 (30) ³
100-250 ft (30.5-76 m)	1 (45) ³	2 (35) ³	2 (35) ³	3 (30) ³	3 (30) ³	3 (30) ³
251-300 ft (77-91 m)	2/0 (70) ³	1/0 (55) ³	1/0 (55) ³	1 (45) ³	1 (45) ³	2 (35) ³
301-350 ft (92-106 m)	3/0 (85) ³	2/0 (70) ³	2/0 (70) ³	1/0 (55) ³	1/0 (55) ³	1 (45) ³
351-400 ft (107-122 m)	3/0 (85) ³	3/0 (85) ³	2/0 (70) ³	2/0 (70) ³	1/0 (55) ³	1/0 (55) ³
Sub-Feeder Length (A1 to PDU)						
0-40 ft (0-12 m)	1 (45) ²	2 (35) ²	2 (35) ²	3 (30) ²	3 (30) ²	3 (30) ²

Notes:

- 1.) The feeder table above is based on the use of copper wire, rated 75C and run in steel conduit. Wire size is in AWG (mm²). Ampacity is determined in accordance with the National Electric Code (NFPA 70), Table 310-16 (1999).
- 2.) The wire size shown is the minimum allowable for the specified over-current protection rating.
- 3.) The feeder is sized to contribute 2.4% maximum regulation.
- 4.) A minimum 1/0 (55 mm²) ground wire is required.
- 5.) A neutral wire is not required and, if present, is not used.
- 6.) The Primary Disconnect (A1) must have provision for tagout/lockout. [Figure 9-7, on page 89](#), shows the primary power disconnect (A1 - P/N R4502AD) using a combination fusible disconnect and magnetic contactor.

Table 8-1 Facility Power Requirements

If the LightSpeed 4.X system must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment which use rapid film changers should be avoided. These changers use a large number of high powered, closely spaced exposures which may coincide with a CT scan and produce image artifacts.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to LightSpeed 4.X system meets all the requirements.

Power Systems for X-Ray Rooms, Direction 46-013833, is available for additional information on power requirements for X-ray systems.

Section 4.0

Uninterruptable Power Supplies (UPS)

Uninterruptable Power Supplies (UPS) are recommended for areas or sites with known power issues. Consult your local power provider for power quality data in your area. UPS is standard equipment on all mobile units. Filter and surge protectors are not needed with LightSpeed 4.X systems.

Section 5.0

Power Audit

A site power audit is required for the LightSpeed family of products. This site power audit can be arranged with the GE Power Quality team, or through your sales person.

A sample audit form is located in the back of Chapter 2.

Section 6.0

Power Source Monitoring

Note:

Perform a power audit first.

A power line analyzer should be used to check the proposed LightSpeed 4.X system site power for average line voltage, surges, sags, reclosures, impulses, frequency and microcuts. A period that includes two weekends should be used to simulate several days of normal use. Analysis of the data and site history of any previous power problems with other X-ray systems or computer installations should be reviewed with your Power and Ground representative. Verify “brown-out” (low voltage) conditions, which may occur during summer months, will not exceed the allowable range shown in [Table 8-1](#).

Section 7.0 Ground System

The LightSpeed 4.X system has been designed to use an equal potential grounding system. The required ground system is shown in [Figure 8-1](#). There are three primary grounding points:

- A system power ground point located in the PM (Power Module).
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

All exposed metal surfaces in patient vicinity are grounded to the reference ground point.

For additional information, refer to Electrical Safety Equipment, Direction **46-014505**.

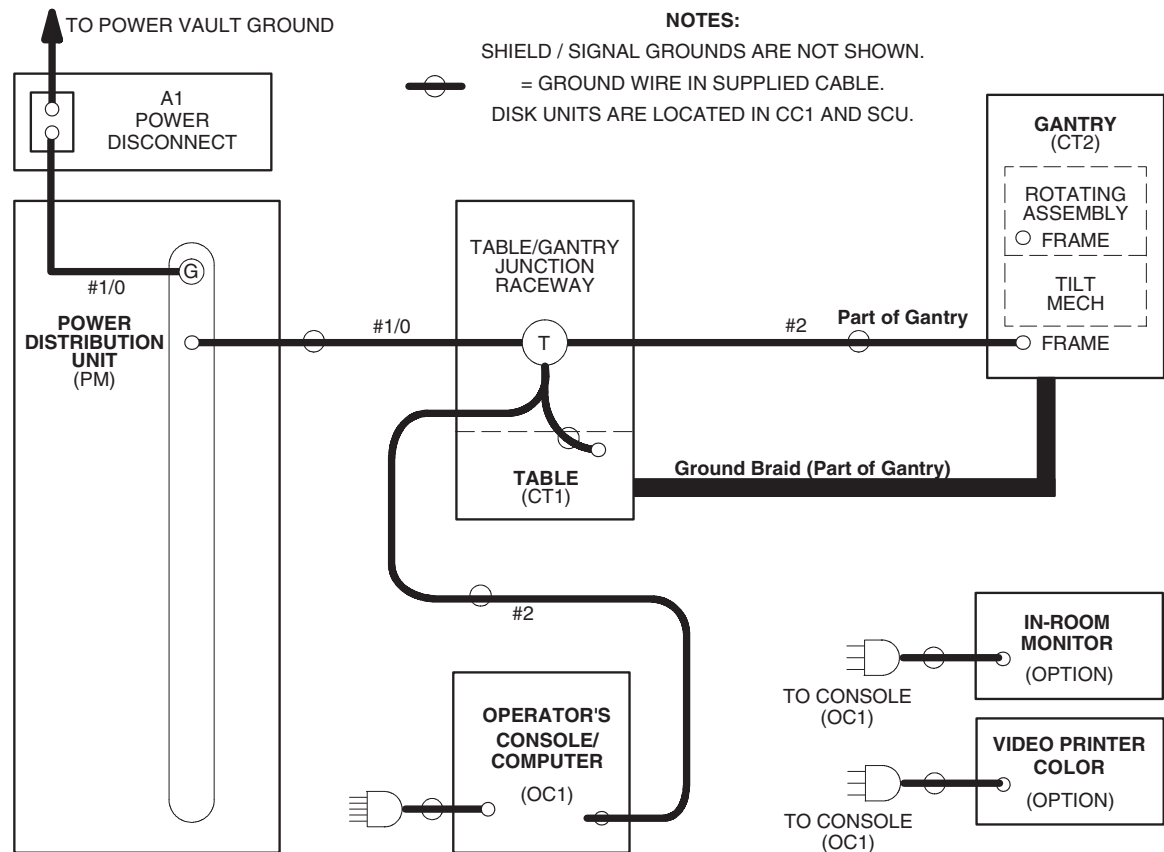


Figure 8-1 System Ground Map

Chapter 9

Interconnection Data

Section 1.0 Introduction

Figure 9-3 shows interconnection runs for a 50/60 Hz system.

Table 9-1 shows component designators for supplied equipment and options and wall power outlets.

Table 9-4 lists customer-installed wiring and supplied cables. Actual length of each run is less than the length of supplied cables to allow for routing inside equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.

Table 9-2 and Table 9-3 list details for connection to LightSpeed 1.3 equipment, using standard (short) length and non-standard (long) length cables, respectively. Details are listed for the following types of runs as appropriate:

- Flush-floor duct
- Computer floor
- Through-wall bushing
- Junction box
- Surface floor duct
- Through-floor duct
- Wall duct
- Conduit

Need for additional junction boxes is minimized by use of either a cable raceway system or a raised computer floor. LightSpeed 1.3 systems use prefabricated cables with large plugs. Therefore, conduit or pipe is not recommended for cable runs.

Section 2.0 Component Designators

DESIGNATOR	APPLIES TO	SOURCE
A1	Primary power disconnect	Contractor supplied
CT1	Patient table	System
CT2	Gantry	System
ITL	InSite telephone lines	Contractor supplied
LP	Line printer	Option
OC1	Operator's console/computer	System
PM	Power distribution unit	System
SEO	System emergency off	Contractor supplied
SM	Slave monitor	Option
WL	"X-ray on" warning light	Contractor supplied
DS	Door Interlock Switch	Contractor supplied
XCVR	Ethernet transceiver	System

Table 9-1 Component Designators

Section 3.0 Interconnect Runs, Wiring and Cables

3.1 GEMS Supplied

3.1.1 Short Cables

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flammability Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
001	60 (55)	18.3 (16.76)	2335376-2	Power - PDU to Console	2587	FT-4	600	120VAC (1φ)	90	12.3 (.483)	4	10	56.4 (2.22) Dia
002	43 (35)	13.2 (10.67)	2275844-3	Ground - PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
003	20 (12)	6.1 (3.66)	2267642-2	HVDC - PDU to Gantry	2587	FT-4	600	+ & - 350VDC	90	19 (.751)	2	6	19.8 (.78) Dia
004	20 (15)	6.1 (4.57)	2266880-2	Power - PDU to Gantry	2587	FT-4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
005	20 (13)	6.1 (3.96)	2267644-2	HVAC - PDU to Gantry	Flexible Motor Supply Cable	FT-4 TC	1000	440Y/254	90	15.3 (.604)	4	12	16.8 (.66) Dia
006	60 (52)	18.3 (15.85)	2275844-4	Ground - PDU to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	20 (12)	6.1 (3.66)	2266014-2	Signal - PDU to Gantry STC	UL	FT-4	300	<30VDC	80	13.3 (.525)	37	22	20 x 75 (.78 x 2.95) 20 x 51 (.79 x 2.01)
101	60 (52)	18.3 (15.85)	2271060-2	Signal - Console to Gantry STC	UL	FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	60 (55)	18.3 (16.76)	2266887-3	Signal - LAN Console to Gantry	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (.234)	1	19	15 (.59) Dia
103	60 (55)	18.3 (16.76)	2117848-6	Fiber Optic - Console to Gantry			NA	NA			1	NA	10 (.39) Dia

Table 9-2 GEMS Supplied Cables (Short Run) - UL Information

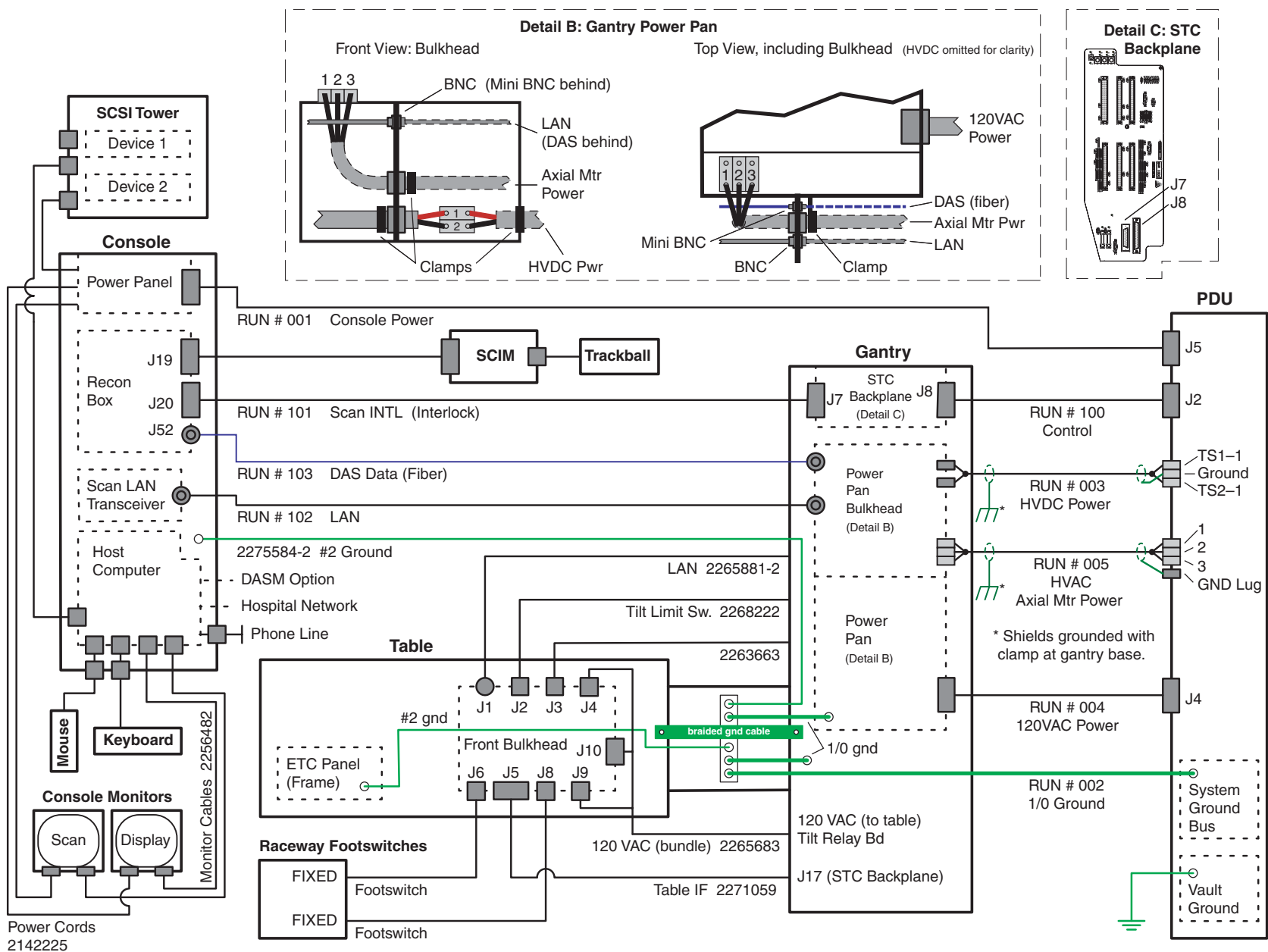


Figure 9-1 System Interconnect Diagram

rev. 03/11/03

3.1.2 Long Cables

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flammability Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
001	80 (75)	24.5 (22.86)	2335376*	Power - PDU to Console	2587	FT-4	600	120 VAC (1φ)	90	12.3 (.483)	4	10	56.4 (2.22) Dia
002	63 (55)	19.3 (16.76)	2275844	Ground - PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
003	63 (55)	19.3 (16.76)	2267642	HVDC - PDU to Gantry	2587	FT-4	600	+ & - 350VDC	90	19 (.751)	2	6	19.8 (.78) Dia
004	60 (55)	18.5 (16.76)	2266880	Power - PDU to Gantry	2587	FT-4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
005	62.5 (55)	19 (16.76)	2267644	HVAC - PDU to Gantry	Flexible Motor Supply Cable	FT-4 TC	1000	440Y/254	90	15.3 (.604)	4	12	16.8 (.66) Dia
006	83 (75)	25.5 (22.86)	2275844-2	Ground - PDU to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	63 (55)	19.3 (16.76)	2266014	Signal - PDU to Gantry STC	UL	FT-4	300	<30VDC	80	13.3 (.525)	37	22	20 x 75 (.78 x 2.95) 20 x 51 (.79 x 2.01)
101	83 (75)	25.5 (22.86)	2271060	Signal - Console to Gantry STC	UL	FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24.3 (22.86)	2266887	Signal - LAN Console to Gantry	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (.234)	1	19	15 (.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic - Console to Gantry			NA	NA			1	NA	10 (.39) Dia

* Cable 2335376, shipped with the Console, is the proper cable for this application. Cable 2266879, shipped as part of the cable kit, should NOT be used.

Table 9-3 GEMS Supplied Cables (Long Run) - UL Information

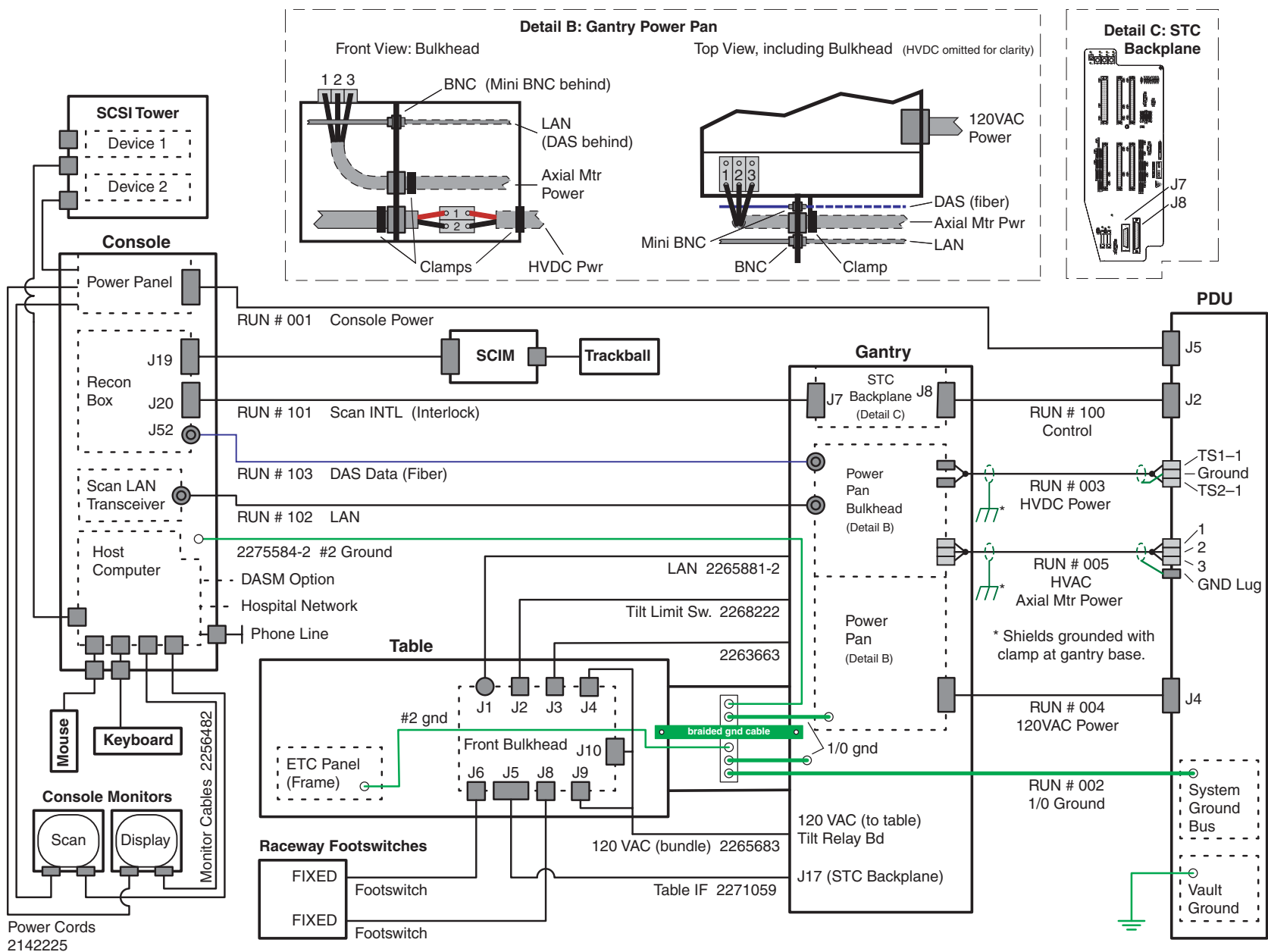


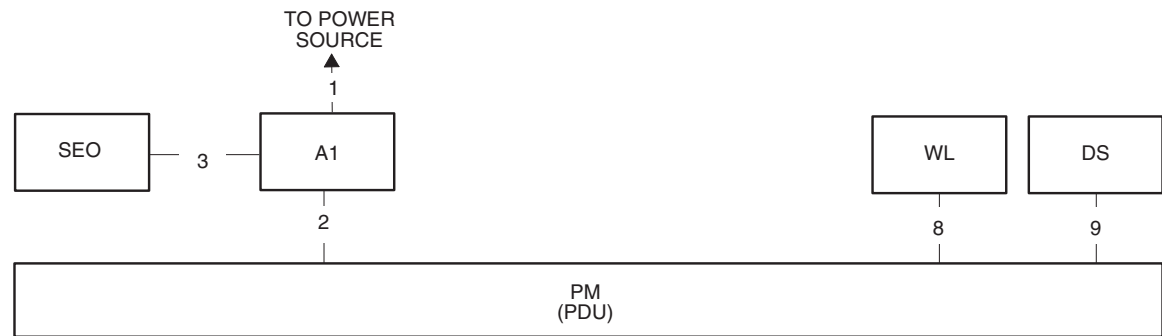
Figure 9-2 System Interconnect Diagram

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3.2 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M)	
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 8 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2-1,2,3,4							
RUN NO. 9 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							
*	REFER TO FOR AWG (MM2) WIRE SIZES.								

Table 9-4 Runs 1, 2, 3, 8 and 9 Connections



NOTES:

- 1) Used for remote diagnostics - Option
- 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
- 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

Only one phone connection is required for the system.

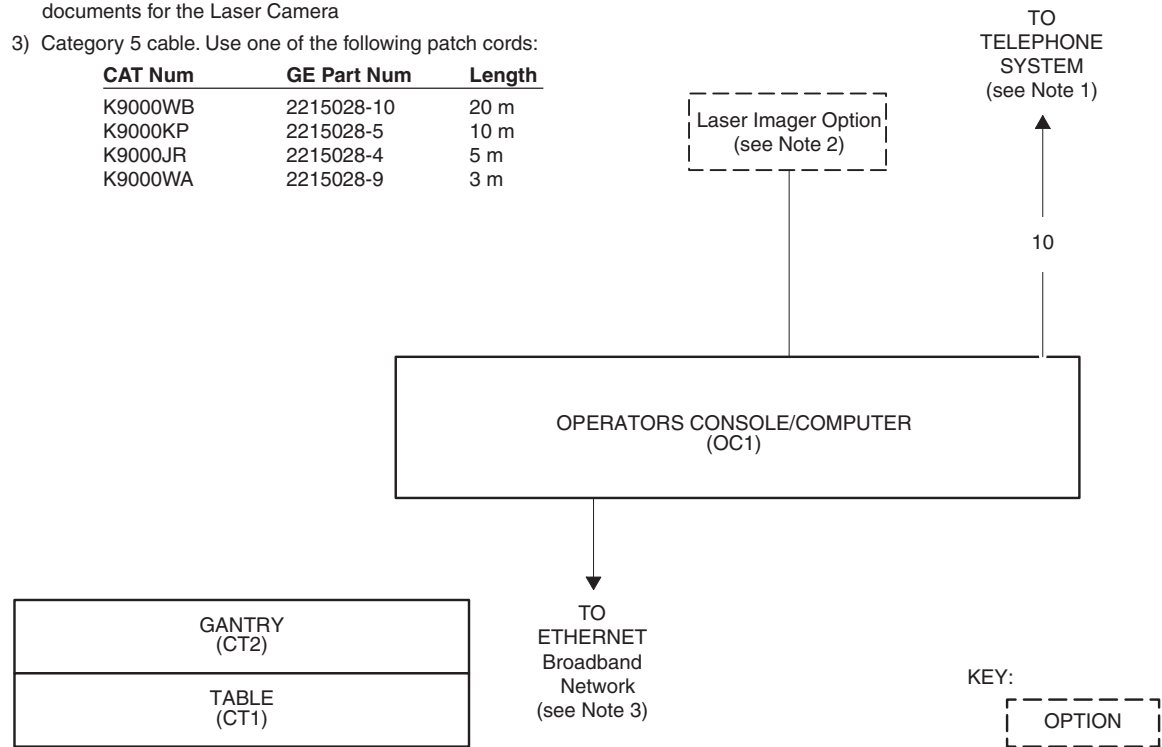


Figure 9-3 Interconnection Runs

Section 4.0 Contractor Supplied Components

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY CUSTOMER CONTRACTOR	USA VENDOR / CAT NO. GE CATALOG
A1 (60 Hz)	Main Disconnect Control (MDC)	480VAC, Surface or Flush Mount, On/Off Control	R4502AD - Main Disconnect Control, 480VAC, Surface Mount with Flush Mount Kit included, two remote Push Button Switches.
A1 (50 Hz) (380V)	Fusible Disconnect and Magnetic Contactor	3 Pole, 380V, Magnetic Combination Contactor. With control trans former for 220V 50 Hz contactor coil, momentary contact start-stop push-button station, indicator light, multiple neutral block, multiple ground block and 160 amp, dual element fuses.	Source Locally
ITL	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by customer.	
	System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area.	
Network Connection		Cat5	

Table 9-5 Contractor-Supplied Components

Section 5.0

UPS Interconnect

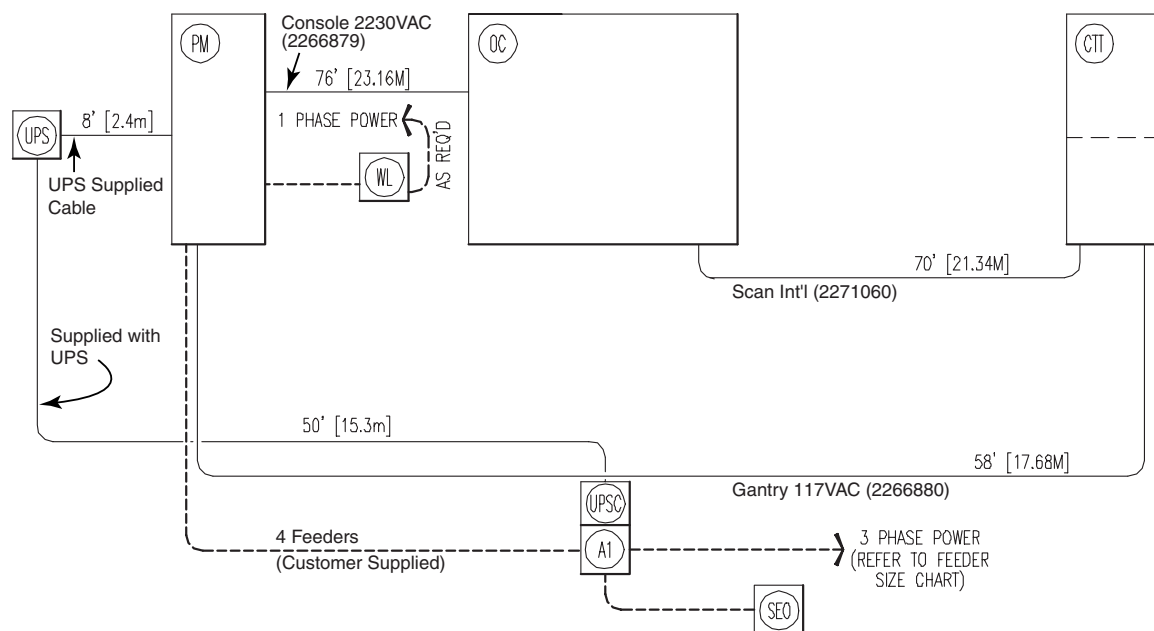


Figure 9-4 Typical UPS Interconnect



Figure 9-5 Typical UPS

CT Systems - Partial UPS Back-up Options						
System Type	PDU Type & Model #	Max.Mom. kVa Rating	Recommended Main Disconnect (A1) Cat #			Optional Partial UPS Kit Cat #
			Europe & Asia	North America		
			380-400V	420V	440V	
HSA-RP	Global PDU 2113764-34	90 kVa	Generic Local Purchase (110A) LOTO Capability Required	Generic Local Purchase (100A) LOTO Capability Required	E4502AD (90A)	B7999PS (incl. 12.5kVa UPS, 30A hardware kit & Aux. UPS Control Panel) (Documentation updates required to support NGPDU-2)
CT _I & LS Qx _I (LS 1.x series)	CRPDU 2133533					
LS Qx _I (LS 2.x series) LS Plus & LS Ultra (LS 3.x series) LS 16 (LS 4.x series)	CT Compact PDU 2269902 or NGPDU-2 2326492-2					
Mobile LS Qx _I (LS 2.x series) LS Plus & LS Ultra (LS 3.x series) LS 16 (LS 4.x series)	CT Compact PDU 2269902 with vibration kit	90 kVa	B7999PY (Incl. Vibration hardened 12.5kVa UPS, 30A hardware kit & Vibration hardened A1 Disconnect Panel (110A) with integrated UPS Control)			B7999PP (incl. 12.5kVa UPS & 40A hardware kit)
LS 16 (LS 5.x series)	NGPDU-3 2326492-3	90 kVa	E4502AC (110A) (incl. Auto Restart & Integrated UPS Control)	E4502AB (90A) (incl. Auto Restart & Integrated UPS Control)	NOTE: REQUIRES one of the A1 Panels shown at left	
LS Pro 16 (LS 5.x series) 100kW, 80kW & RT	NGPDU 2326492	150 kVa	E4502AF (150A) (incl. Auto Restart & Integrated UPS Control)	E4502AE (125A) (incl. Auto Restart & Integrated UPS Control)		

6.1 Primary Power Disconnect

Figure 9-6 Primary Power Disconnect (A1) – Fusible Disconnect and Magnetic Contactor

Figure 9-7 Primary Power Disconnect (A1)

6.2 Scan Room Warning Light & Door Interlock

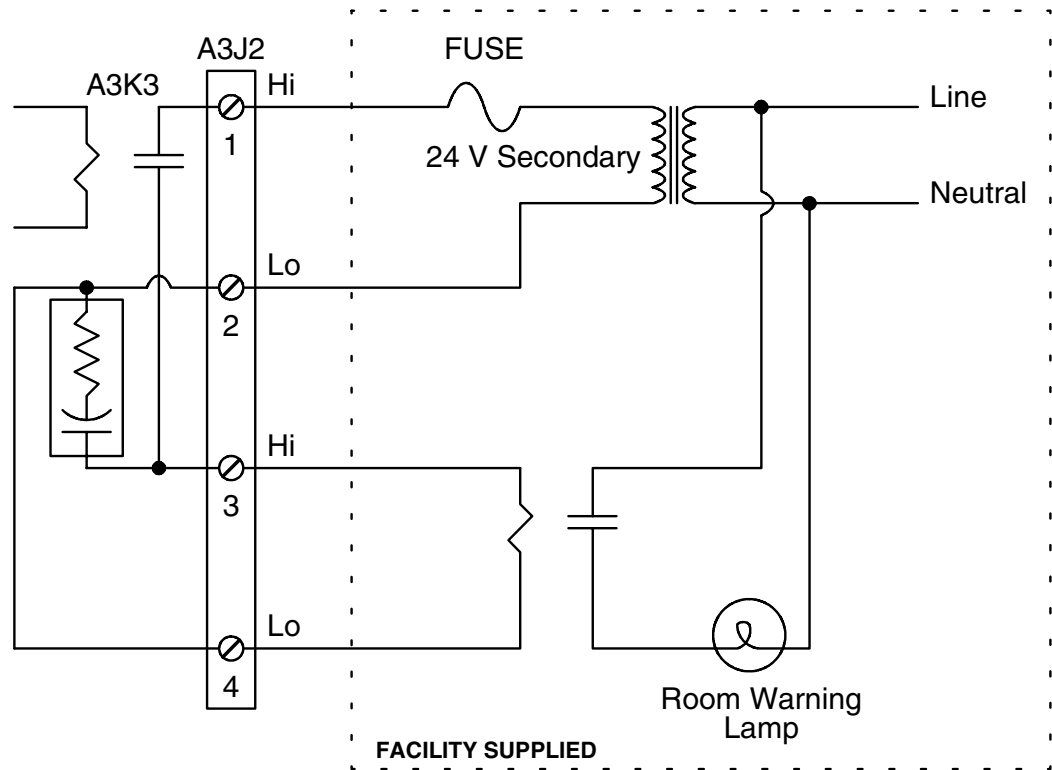


Figure 9-8 Typical Customer Supplied Scan Room – Warning Light Connection to PM

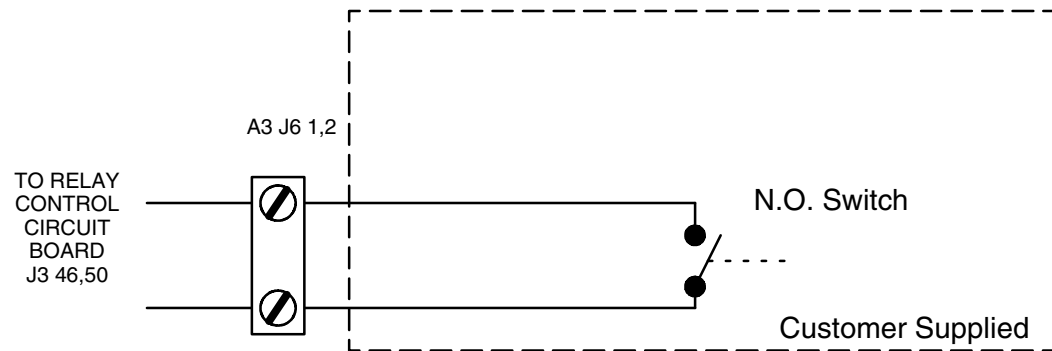


Figure 9-9 Typical Door Interlock

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare.
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com

